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Tracing Semantic Drift on Social Media: A Graph-Based Approach to Meaning Shift in Online Discourse

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Abstract

This dissertation investigates how the meaning of a word can change over time in online environments. This meaning change can be characterised as sematic drift, where the meaning of a word evolves away from its original understanding over a period of time. The specific type of semantic drift that is focused on within this dissertation is pejorative drift. This is where a word's meaning has evolved to become more negative.

The investigation in this study is grounded in sociolinguistic theory that is informed by the concept of indexicality of word meaning. It develops a hybrid computational methodology by combining sentiment analysis, word embeddings and graph-based modelling to effectively detect and visualise semantic drift.

Video metadata (e.g. video titles, video description and top-level comments) was collected from YouTube using the Data API. This data was process and analysed using natural language processing toolkits to compute sentiment scores and track the distributional changes of a word's meaning through cosine similarity word embedding comparisons. A semantic graph database was constructed in Neo4j to visualise the relational drift of a word's meaning, showing the co-occurrence patterns between users and their comments across time.

The results from this investigation showed a clear declination of sematic trend for the chosen word 'woke' while also confirming this drift through measurable shifts is distributional representations. This confirmed the hypothesis that the word 'woke' had undergone pejoration. The Neo4j graph outputs revealed a changing contextual environment for the target word that further supports how the word had undergone recontextualisations of its semantic neighbourhood. However, due to some technical constraints, a full relational comparison could not be made for all the years.

The study demonstrated the value of computational approaches to capturing as part of ongoing theory-based language analysis. It also highlighted the importance of context in shaping meaning as language terms are expansively used within dynamic online communities.

Declaration

I have read and understood the College and Departmental statements and guidelines concerning plagiarism. I declare that:

This submission is entirely my own original work.

Wherever published, unpublished, printed, electronic or other information sources have been used as a contribution or component of this work, these are explicitly, clearly and individually acknowledged by appropriate use of quotation marks, citations, references and statements in the text.

It is 12759 words in length.

Any use of generative AI has been clearly acknowledged in my work and is permitted under the assignment's categorisation according to UCL's AI guidelines

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1 Introduction

1.1 Research Background

In this project, I will be investigating how the meaning of words shifts over time in online environments. The core element being investigated is semantic drift, specifically the type of semantic change that can occur when a word is adopted across varying socially stratified communities (e.g. word pejoration that causes a word's meaning to gradually obtain a more negative sense).

The primary objective of this project is to identify and detect this semantic drift and visualise the change with graph-based models. The study engages with a variation of sociolinguistics practises, with focus in the research areas of systematic variation of language as it spreads in between different speech communities.

This project will not examine any arbitrary shift or complete meaning replacements of a word, as this is not fitting with sematic drift. However, it will be focusing on semantic drift as the evolution of a words meaning without discarding original sense of the word. Therefore, I will attempt to locate examples of drift for a word where there has been a collective change of how the word is understood within social discourse.

The study is focused on language used on YouTube. YouTube is an online video sharing platform that prioritises user generated content to be shared globally, where users can engage directly with other's content by liking, sharing and leaving comments under videos and community posts. The video titles created by the Youtuber (user that creates YouTube videos) and the comments left by viewers (users that watch YouTube videos) under said videos will be collected and analysed using keyword-based searches to trace how the meaning of a word is influenced by the sociocultural context in the digital interactions in which it is used.

1.2 Motivation

This project is motivated by a desire to understand how public discourse can shape the way that language is used online. The pejorative semantic drift of language is often amplified in digital platforms due to its unique ability to create rapidly amplified echo chambers that can drive discourse in polarised directions. By analysing these shifts computationally, I aim to uncover a deeper sensitivity to the sociopolitical undercurrents of everyday language.

1.3 Research Questions

This study aims to explore the capabilities of computational approaches to detecting and visualising semantic drift of language used in digital environments. The study explores varying graph-based approaches to visualise the evolution of a word's meaning over time, specifically the pejorative declination that occurs causing a word to become more negative in meaning. The approaches developed in this study aim to provide a linguistic framework for natural language analysis that is data driven, scalable and reproducible.

This study intends to investigate these research questions:

main research question that I aim to answer in this investigation is the following:

1. How can computational method for sentiment analysis be used to track semantic drift in social media discourse?
2. Are graph-based representations able to accurately illustrate the pejorative drift in the target word?

1.4 Overview of Methodology

The purpose of this project is to track how the meaning of a word can change over time across different communities. The overall aim of the study is producing quantified graph-based visualisations of semantic drift and build a semantic graph in Neo4j to visualise word evolution from year to year: with some insights into how the context changes over time.

The project is focused on quantitative research processes with some qualitative data analysis pertaining sociolinguist research that may provide more in-depth insights into the why language evolves in that way. The study adopts the analysis approach that critically examines data to identify patterns and meaningful insights through data visualisation techniques.

Data will be collected from YouTube Data API using keyword-based searches to identify text (titles and comments) that include the targeted word of the study. Data will be collected on users active from the years 2010 to 2025. The individual consent of users does not have to be obtained due to the implicit ongoing consent based on users' agreement with YouTube's terms to abandoning anonymity when posting publicly. However, user metadata will be anonymised under privacy regulations such as GDPR.

Sentiment analysis graphs will be constructed for different types of graph queries (e.g. sentiment scoring and co-occurrence calculations) to show how a word's meaning has shifted away from its original sense. For the graph database built in Neo4j, each node will represent a sentiment score, context (e.g. adjacent words), with each edge encoding any co-occurrence. The cypher function of Neo4j should allow for efficient data querying of the graph to detect semantic changes.

Embeddings will be used to perform similarity comparisons to extract meaningful representations beyond string comparison, based on cosine relationships between nodes.

The computational findings from the study will then be combined with the interpretive readings of shift from the selected time periods to identify real-world events or ideological trends that may be cause for the phenomenon.

1.5 Dissertation Structure

This dissertation consists of five sections. The first section introduces background of the study, the motivation behind it, the research questions and an overview of the methodology.

The second section reviews literature related to the study of semantic drift and the theoretical frameworks that can analyse it, how semantic drift can occur in social media, and the computation approaches of detecting and visualising semantic drift.

The third section details the methodology of the hybrid computational approaches developed in this investigation for detecting and visualising semantic drift in online discourse. Highlighting the data collection and data processing pipeline created to generated graph-based visualisation of the semantic drift of the target word.

The fourth section presents the results from the graph-based visualisations of semantic drift along with analysis into the data that was obtained. This section also includes visuals form the Neo4j databased created.

The final section five evaluates the computational approaches developed in this study.

2 Literature Review

The main topics that will be discussed in this literature review are the theoretical frameworks used to analyse semantic drift, how semantic drift can be observed in social media and the computational approaches to detecting and visualising drift.

The first section will provide a definition for the type of semantic drift being investigated and focus on the theatrical models from ethnomethodology and semeiotics that have been traditionally used to examine how meaning is shaped through context and interaction.

The second section will contextualised semantic drift within digital environments and explore how meme culture and sociopolitical discourse can drive meaning change online. A case study 'woke' and its word evolution will be the focus in this section.

The third section will review and evaluate the computational methods that can be used to trace and visualise semantic drift. These methods include distributional semantics, sentiment analysis and graph-based models.

2.1 Semantic Drift

2.1.1 Introduction to Semantic Drift

As society evolves, so too does the human language; it adapts to reflect the cultural transformations, ideological shifts and the reframing of social meanings attributed to words. At the same time, it must keep pace with the rapid dynamic of discourse that is enacted online. This phenomenon is defined as *semantic drift*, where the meaning of a word evolves over time.

Semantic drift is highlighted throughout (Liberman, 2012) investigation into indexical expressions, which recognises the lability of word meaning and refers to the idea that meaning is not fixed but inherently unstable. This instability is reflected in the way that meaning is not derived solely from individual words in a conversation, but can emerge through the course of interactions, allowing for word meanings to evolve with time. The concept of indexical expressions will be explored deeper in **Section** 2.1.3 Theoretical Frameworks Explaining Semantic Drift.

(Liberman, 2012)'s understanding of semantic drift highlights its dependency on context. Meaning may initially be grounded in prior knowledge but is also shaped and reconfigured through the progression of a conversation. Through these dynamics, we can observe various forms of semantic drift, each characterised by how much the meaning has changed and how it is interpreted. The next section examines the specific features that separate the different types of shifts.

2.1.2 Types of Semantic Drift

The evolution of a word's meaning can be gradual or abrupt; in both cases, the shift can be significant enough that the new meaning substantially diverges from its original usage. So much to the extent that the word's initial intent or function is obscured or lost.
To illustrate this point, (Wijaya & Yeniterzi, 2011) cites awful[1] as a classic example of how a word's meaning can undergo semantic drift. This word originally meant 'awe-inspiring' but is now used informally to describe something as 'very bad'.

Widening	*bird*	'young bird' > 'any kind of bird'
Narrowing	*meat*	'food' > 'edible flesh'
Amelioration	*knight*	'servant boy' > 'nobleman'
Pejoration	*awful*	'inspiring wonder' > 'terribly bad'

Table 1*: Semantic Change examples*

Source*: Table 1.2* (Dubossarsky, 2018)

As shown in **Table 1** the word 'awful' has undergone *pejoration*, where the meaning of a word has changed to become more negative.

Some of the other recurring types of semantic shift involve, changes in the connotation of meaning to become more positive (*amelioration*), and changes in scope to be inclusive (*widening,* also known as *broadening*) or more restrictive (*narrowing*).

Pejoration is a common semantic change that can be observed in modern discourse, as it can reflect the regular shifting cultural or political ideologies within society. It is best visible in online platforms as the clear interactions between users showcases how words can take on a

[1] https://dictionary.cambridge.org/dictionary/english/awful

new negative contextualisation, often through sarcastic or oppositional contexts that gradually alter the perceived meaning among users.

Building on this observation, this dissertation aims to focus specifically on pejoration as a form of semantic shift in online discourse. For example, the word 'woke' provides an illustrative case of pejoration in online discourse, as we have seen its' meaning undergo significant shift towards a more negative or mocking contextualisation over time.

2.1.3 Theoretical Frameworks Explaining Semantic Drift

Semantic drift is a topic of Ethnomethodology, a method of sociolinguistic analysis that examines how language used within everyday conversations dictate and determine the understanding of the interactions. (Liberman, 2012) labels the phenomenon under the title "indexical expression", summarising how the meaning of a word becomes intelligible only through its use in a specific conversational context. The meaning of a word is not fixed in advance but is locally constructed and shaped by the people participating within the conversation, negotiating sense as dialogue unfolds (Garfinkel, 1967) (Garfinkel & Sacks, 1970).

(Liberman, 1999) notes that "words and texts do not possess their meanings unequivocally in themselves; rather, the occasioned discourse taken as a whole carries the sense". This supports the idea that meaning is obtained through the collaboration of the interlocutors[2]. Meaning is shaped by the contingencies of the interactions, and the efforts made by the speaker and listener to achieve mutual understanding of said interaction.

The study of semiotics based on the discovery that words should be treated as part of a shifting system, as opposed to being a part of a static container of meaning. Thus, permitting language to be interpreted with an open sense to allow for different understandings to be applied, with sense and intelligibility being distributed and renegotiated across the interactions between people (Liberman, 1999).

Saussure's semiotics establishes language as an integrated system of signifying elements *syntagmata*, arguing that the valuation of a word is derived by its oppositional relationships

[2] https://dictionary.cambridge.org/dictionary/english/interlocutors

between the other elements in the system. Therefore “all meaning is derivative” (Liberman, 1999) and not intrinsic, always open to reinterpretation.

Ethnomethodology brings this theoretical insight into practise by showing how both the speakers and listeners collaboratively construct meaning in real-time. (Liberman, 2012) and (Liberman, 1999) emphasises that meaning is not a fixed property of individual words but can emerge through the collaborative activity of the interlocutors reflexively responding to the unfolding context. (Liberman, 2012) demonstrates this with his example depicting the negotiation of meaning for the word *five*.

In **Illustration 1** (Liberman, 2012), meaning has emerged mid-conversation, shaped by how the word *five* was first interpreted and then clarified once the context was understood. Here we are experiencing the phenomenon of equivocality, that observes the possibility of words existing under several different meanings. In this example, we see how the role of the listeners is as important to the role of the speaker to establish context.

Illustration 1

KL Do you have any apple juice?
IN Yes.
KL How much is it?
IN Six rupees.
KL I’ll take five.
IL No, I’m sorry. Six rupees.
KL Five, I’ll take five [holds up hand with fingers].
IN Mm, Alright. [3.0 pause]
KL Here, [Hands the shop keeper three 10-rupee notes]. Five [displaying five fingers].
One. Two. Three. Four. Five. [Motions to cans on shelf.]
IN Oh. [Retrieves five cans from shelf.]
KL One more. [Holds up one finger.]
IN [First unwilling, then retrieves one more.]
KL Thank you.
IN O.K.

***Source**: Illustration 1 (Liberman, 2012)*

We see semantic drift unfold in everyday language use, not limited to just a historical process for language evolution. Shifts in natural language can emerge from small-scale social or political contexts, where meaning may be transformed through the mutual confirmation processes involved in the multiple levels of subconscious and conscious language learning done throughout everyday conversations.

2.1.5 Contextualising Drift in Digital Language Use

Social media platforms provide distinct linguistic environments for users to communicate and interact with one another. These environments have their own interactional norms, shaped by technical affordances (e.g. character limitations) and community preferences that emerge within each platform. For example, on the social media platform X (formally known as Twitter), users' tweets are restricted to 280 characters. As a result of the linguistic constraints like the ones mentioned above, users are often compelled to abbreviate phrases and creatively reframe speech to more efficiently convey their meaning. This practise reiterates (Heller, Sari, & Pujolar, 2024) claim of meaning being a "constitutive product of social interaction that is embedded in power relations".

These conditions can give rise to what can be understood as platform-based digital dialects. Mirroring how offline dialects are formed as language evolves independently in groups isolated due to social differentiation and or their distinct communicative needs. An individual's language use can indicate to the specific social context and groups in which they reside in, including the online spaces that they frequent.

We see reflected in this dynamic how meaning can be shaped by specific social groups and community norms. For all online spaces, each have their own linguistic conventions that make it difficult to establish a unified digital language. The fragmentation of language across these online platforms causes users to be dependent on context and usage patterns to achieve mutual intelligibility. Social media spaces are environments that fosters users to constantly contest and negotiate meaning through discourse. Such environments thrive by allowing the coexistence of varying linguistic communication styles, so the users are expected to extract meaning dynamically from context and interaction. Supporting Lieberman's argument on meaning not being encoded in words but negotiated as context unfolds.

Consequently, semantic drift in online discourse can be seen as occurring horizontally, across distinct online communities and social groups within the same generation. As well as occurring vertically with language evolving over time. The online social media environment offers a great source for observing how meaning is not only temporally fluid but can be socially distributed. Communication has become a powerful tool for shaping one's identity, and through analysing language we can identify the broader social patterns that emerge as online language use changes.

The pragmatic and ethnomethodological perspectives show that meaning is not inherently fixed across all contexts. Lieberman's notion of "indexical expression" highlights how meaning is constructed through the interaction between the speaker and the listener. It is the speaker's intent and the listener's interpretation that shapes how context is understood through an evolving conversation. This supports the view that semantic drift is not limited to a historical process but can occur locally with everyday conversations. These dynamics are especially visible in social media environments where meaning is often renegotiated in real time through replies and comments. Examples exist where singular terms or phrases can shift in connotation depending on the group that interacts with it, illustrating how online platforms can become key locations for observing contextual drift in action.

2.2 Semantic Drift on Social Media

2.2.1 Language use in Social Media

The expansion of social media has created a unique environment for observing semantic drift. These online platforms have developed a digital space for rapid paced communications of thoughts and ideas through a typically informal register, transforming how meaning can evolve over varying digital contexts.

Language is exchanged across diverse demographics, evolving within digital spaces where individuals can come together to share ideas and interact across community boundaries. Social media platforms are designed for globalised user interactions to enable people to connect through comments, shared posts, likes, and engagement.

Language evolves through these shared digital practises, enabling the emergence of new meaning that is influenced by these online communities. This fosters a unique environment

for linguistic change that is shaped by the continuous communication and mixing of cultural ideas in online spaces.

2.2.2 Drivers of Semantic Drift in Social Media

Digital communication is not only shaped by the users, but also the interactional norms that mutate language to be more effective and efficient. The language used online is in constant negotiation of meaning to keep up with the everchanging trends of culture, memes and algorithmic visibility that shape how interactions are to be understood at any given time. We see this demonstrated in (Mullick, et al., 2018)'s investigation into *topic drift* on social media.

(Mullick, et al., 2018) discusses examples where the topic flow has drifted outwards from the initialised context. In one example, the shift is prompted by a change in the user's focus. The conversation started with discussing Tspiras and Greek's economic situation to invoking Germany's historical role in helping to reconstruct Europe. This is a clear example of topic drift, where the shift has recontextualised the conversation through geopolitical inquiry. We see reflected how discourse on social media can evolve in response to shifts in contextual framing and user intent.

User-generated content on social media frequently exhibits contextual shift in conversation because of misinterpretations of user intent or context. However, this shift can also be the result of the intentional reshaping of words through acts of irony, oppositions, and or appropriation. (Shen, 2024) argues that semantic change on online platforms like Twitter and Reddit are deeply reflective of the wider societal dynamics that govern its users.

In recent years, we have seen the strategic recontextualization of certain words and terms on online platforms, wherein the original meaning has been stripped and reassigned to more negative characteristics to mock or delegitimise the communities that once used them. This form of real time semantic drift demonstrates how language can be weaponised for political or ideological positioning, underscoring the inherently unstable and negotiated nature of meaning within online discourse.

2.2.3 The Sociopolitical Dimension of Drift

The article "How the word 'woke' was hijacked to silence people of colour" (Morris, 2020) offers a case study of how the word 'woke' underwent semantic drift on online platforms. The word 'woke' was first used in the 1940s as a political term to indicate whether someone was "awake to issues of social justice and racial justice" (Morris, 2020). It then had a later resurgence in the mid 2010s during the rise of 'Black Lives Matter', gaining rapid visibility via hashtags and viral discourse. However, as its usage spread beyond its original community, 'woke' became increasingly recontextualised and weaponised by popular commentators and media outlets to mock progressive attitudes (Morris, 2020). This example illustrates an intentional act of semantic drift in the pejorative sense through acts of political subversion.

Additionally, the viral uptake of the word helped to dilute the meaning from its original intention, as meme culture can accelerate semantic erosion, removing it from its initial context (Morris, 2020).

(Mullick, et al., 2018)'s findings suggest that the "root users" who initiate these shifts are often highly opinionated and express a heightened sense of emotion. Semantic drift tends to correlate with increased engagement, especially when aligned with controversial or viral content. This demonstrates how meaning change is not only limited to natural linguistic evolution but can be shaped by user behaviour and the engaging content that is capable capturing widespread attention.

2.2.4 Types of Semantic Drift Observed Online

We see certain words, as they are increasingly used in online spaces, expand in scope and meaning. The words listed in **Table 2** are particularly susceptible to drift due to the specific cultural and subcultural communities that they are frequently used in.

These communities can be seen as more polarising, so when discourse crosses over into the mainstream it leaves the language vulnerable to misunderstanding or reappropriation. For example, 'woke' began in African American Vernacular English (AAVE) spaces as an alertness to racial injustices but was reappropriated by critics to make fun of socially liberal ideologies (Morris, 2020).

Key Term	Original meaning on social media	Shifted meaning	Type of shift	Example
Woke	AAVE term meaning "socially aware" or "awake" to racial injustice	Used dismissively to mock people seen as performatively progressive or overly politically correct.	Pejoration	"The company is just being woke for PR"
Simp	Someone (usually a man) who does too much for someone they like	Mocking insult for any form of kindness or support shown towards women.	Pejoration	"He donated to her stream? What a simp."
Pick-me	Slang for individuals (often women) who seek male approval by putting down others	Vague insult for people perceived as attention-seeking or lacking self-respect.	Widening pejoration	"Ugh, she's being such a pick-me"
Karen	Common first name	Used online to describe a woman (often white, middle-class) who is entitled, demanding, or exhibits racist behaviour, especially toward service workers or people of colour.	Pejoration	"She really pulled a Karen at the store today."

***Table 2**: Examples of words that have undergone Semantic Drift on Social Media*

It is important to note, that words such as 'woke' and 'simp' have already undergone a semantic change via a connotation change. Where to be 'woke' is a shorted word from 'awoken' and 'simp' is a shortened term for a 'simpleton'.

These examples also illustrate how terms can be fluidly interpreted online. The key terms given in **Table 2** are often deployed online through varying senses. Whether that term is to be understood ironically or critically, it is dependent on the speaker, audience (listener) and the platform to establish the context. Therefore, interpretability must be prioritised.

The meaning of these terms is often dependent on how and where they are used, so it is important that the tools used to analyse semantic drift are able to identify the causes of the change and how it has happened. This is particularly valuable for capturing subcultural uses of words and terms, where meaning is negotiated locally and is changing as it is used by different groups online.

2.2.5 Summary

Social media has accelerated the phenomenon of semantic drift, particularly shifts that are shaped by ideology and the virality of discourse online. As (Mullick, et al., 2018) notes through topic drift, meaning can be recontextualized in real time based on algorithmic visibility and user engagement. We see this exemplified through (Morris, 2020)'s case study on the word 'woke'.

In this example, drift has been politically weaponized to strip the word of its original intent and reframe it through an ironic and mocking sense. Similarly, (Shen, 2024) argues that shifts on platforms such as Twitter and Reddit are often driven by appropriation or oppositional discourse that can reflect wider societal tensions. These examples show how semantic drift online can be reflective of the broader sociopolitical dynamics that exist beyond social media.

Drift can provide a critique to the communities that exist online but can also enable the memeification[3] of language when language crosses into mainstream discourse. Consequently, these examples have underscored the importance of context for understanding the meaning

[3] https://dictionary.cambridge.org/dictionary/english/memeification

around words. Therefore, the need for interpretability is essential to analysing drift in environments where meaning can change fluidly.

2.3 Computational approaches for detecting and visualising semantic drift

2.3.1 Introduction to Computational Approaches

Computational approaches to semantic drift refer to the methods and computational tools used to detect and quantify how word meanings shift over time.

Traditionally, semantic change has been studied using a top-down approach, where theoretical frameworks are used to analyse the meaning of words. This can be best understood as a "close-reading" approach (Tahmasebi, Borin, & Jatowt, 2021), where a few words of a text are selected to then be analysed against linguistic theory. However, this approach prevents patterns from emerging from the raw data alone. This is because it is aimed at only testing specific hypotheses derived from linguistic theory, causing it to be prone to missing any broader patterns that may emerge from the text.
In recent years, the digitization of historical documents has facilitated more advanced data-driven analysis of semantic change. The increased availability of large digital text collections (corpora) and advanced computational tools can allow for more scaled-up research potentials of semantic change, compared to the smaller-scaled traditional approach previously used by linguists.

The computational methods allow researchers to automate the processing of large corpora and detect subtle shifts or patterns that would have been missed if done manually. This dissertation will utilise computational approaches to detect and visualise pejorative shifts in a word's meaning within large scale online discourse.

Paradigm	Analysis Approach	Scale	Common method(s) Used
Traditional	Theory driven	Small	Manual Analysis
Computational	Data driven	Large	Modern NLP and modelling

Table 3*: Comparison of Research Paradigms: Main ways that semantic change is studied*

Adapted from Source*: Table 1.1* (Dubossarsky, 2018)

Table 3 is an adapted summary of the traditional and computational research approaches. The most used computational techniques include Natural Language Processing (NLP) pipelines, Distributional Semantic Modelling (DSM) to represent word meaning and graph-based models (Dubossarsky, 2018).
DSMs use word-embedding methods such as word2vec and GloVe to model word meaning based on co-occurrence patterns. Graph-based approaches focus on representing semantic relationships and how the contextual environment changes over time.

2.3.2 Methods for Detecting Semantic Drift

Distributional semantics and embeddings + Time-based change detection

This dissertation incorporates graph-based representation with distributional elements, relying on semantic networks built using user-generated YouTube data and cosine similarity between vector embeddings of words to detect semantic drift. This hybrid approach allows for the contextual shifts and the structural changes in the meaning of words to be modelled altogether.

The aim of this section is to explore the key computational techniques and methods for detecting and visualising semantic drift. These include graph-based methods, embedding models and visualisation tools.

This section will also examine the suitability of the computational approaches for detecting meaningful drift of used with in the social media context, highlighting any challenges that may occur from the approaches. In this case, meaningful drift is that is significant enough to be considered drift.

Distributional hypothesis (Harris, 1954) argues that the meaning of a word can be extracted from the contexts in which it is used. Vector space models (VSM) are based on NLP tools derived from this hypothesis, using the "local context window around a given word to capture information about the words that co-occur with it" (Dubossarsky, 2018). Otherwise said, it looks at the words that would usually appear near the given word to understand its meaning based on context.

VSMs are essential tools within the computational approach to studying shift. Each word is represented as a single continuous vector in a high-dimensional space. The cosine-distance between vectors represent the degree to which the words are semantically similar. Therefore,

we can use the cosine-distance to estimate the degree of semantic change. A large cosine-distance would mean that the word has undergone a significant shift in meaning while a smaller cosine-distance would suggest greater semantic stability. A representation of the cosine-distance can be seen in **Error! Reference source not found.**. Where $v_{\omega}^{t_0}$ and $v_{\omega}^{t_1}$ are the vector representations of the word ω at two time points, t_0 and t_1, respectively (Dubossarsky, 2018).

$$\Delta\omega^{t_0 \to t_1} = 1 - \frac{v_{\omega}^{t_0} \cdot v_{\omega}^{t_1}}{\| v_{\omega}^{t_0} \| \cdot \| v_{\omega}^{t_1} \|}$$

Equation 1*: Cosine-distance*

Source*: Equation 1* (Dubossarsky, 2018)

The main limitation with VSMs is that each word is represented with a single global vector. According to (Dubossarsky, 2018)**,** this global vector representation is disadvantageous for semantic drift research, because it offers little insight into which senses were involved in the word's meaning change and the elements that have changed.

Polysemy exists in natural language, where multiple meanings can coexist for one word. So VSM collapsing these senses into a single global vector limits semantic drift research, as semantic change can involve the development of new sense (meaning) and the disappearance of others. This global-embedding approach in unable to distinguish semantic shifts by sense.

There are several computational strategies that can be used to investigate and analyse how words are presented within semantic drift over time. Vector space models have been particularly highlighted within (Tahmasebi, Borin, & Jatowt, 2021) as a common representation method to enable a direct comparison of word vectors across different time slices.

However, this method does not account for when semantic change has occurred, or for which changes are significant. Therefore, to directly map the change, the change point detection method (Tahmasebi, Borin, & Jatowt, 2021) can be applied to further expand on the techniques used to observe the shifts in word representation across time.

The change point detection method is used to detect when semantic change has occurred. This method involves the mapping of all time specific embeddings to a shared reference

space/ common timeframe that allows for the changes of a word's contextual environment (i.e. semantic shifts) to be tracked and compared across time.

The key points in which a words representation has changed can be quantified and measured using metrics such as cosine distance. As illustrated in **Error! Reference source not found.**, the angle between a word vector at t_0 and t_1 can provide an interpretable way to capture the amount of change that a word has undergone. In other words, acting as a metric quantifying the degree and trajectory of shift across time. Through this we can isolate the words that are undergoing meaningful change and distinguish them from the terms that remain stable in their representation.

Graph-based and network models

As the volume of online communication on social media increases, so too does the scale of the data infrastructure of the platforms required to support these communications. This data infrastructure not only includes user generated content, but also the complex relational data that underpins the platform's functionality. (Aryono, 2016) proposes that a graph-based approach is best suited to managing these complex relationships by using principles from graph theory to maintain performance during data retrieval. In this framework semantic graphs are composed of nodes and edges. The nodes represent entities (e.g. word or concepts), and the edges represent the relationships that connect them (e.g. contextual or co-occurrence relationships connecting them). Meaning is not derived from the entities themselves, but from the dynamic patterns that emerge from their connection. Graph databases structured through graph theory allow meaning to be stored relationally so that the data can then be efficiently queried over time.

Neo4j has been selected at the database platform to construct and analyse semantic networks. Graph databases like Neo4j are specifically designed with the capability of querying large graphs for real-time traversal of large and complex datasets (Neo4j, 2022). Its property graph model enables the representation of data as nodes and edges that can store rich relationship data, making it particularly suited for tracking semantic drift over time. Neo4j operates using the Cypher graph query language. Furthermore, graph databases are often described as schema optional, providing the flexibility required to represent evolving semantic relationships without imposing rigid data constraints (Guia, Soares, & Bernardino, 2017).

Semantic drift can be detected in Neo4j by modelling the evolution of a word's meaning over different time slices and comparing their structural changes. In graph theory, the neighbourhood of a node is the set of nodes that are adjacent to it. If there are no other nodes are adjacent to it, it is independent (Aryono, 2016). The neighbourhood in a graph-based model can be extracted to observe how relational structures evolve over time. This method allows for semantic drift to be detected relationally, where meaning is derived from the structure of the nodes and relationships. This approach emphasises how meaning is embedded in the dynamic structure of data relationships. Expanding on the focus on the distributional similarity methods of vector space modelling. Thus, enabling shift to be effectively identified through changes sensitive to context across online environments.

2.3.3 Approaches to Visualising Semantic Drift

Semantic network visualisation

Semantic drift is a phenomenon that is difficult to detect from raw data alone. Therefore, we must rely on techniques that can consolidate the data in a way that transforms vague patterns identified in information into meaningful insight. *Visualisation* is emphasised by (Chen & Floridi, 2013) to function as a form of "computer-aided seeing" that can aid in identifying anomalies and trends in how words relate to one another over time.

The task of uncovering semantic relationships would otherwise be difficult to achieve or interpret if the data remained in its raw, unstructured form. The visualisation pipeline model shown in **Figure 1** illustrates the value of visualisation in its ability to enrich and filter semantic information so that it can be mapped and rendered for user interfaces.

Visualisation enables "memory externalisation" (Chen & Floridi, 2013), described as an operation that helps users to form insights by connecting visual patterns with cognitive interpretations. This is especially important for the visualisation of semantic drift, as it allows shifts in meaning to be traced in a way that is interpretable for both expert and non-technical audiences.

Visualisation can be used as a tool to critically explore and analyse semantic drift by bringing visibility to information that is meaningful, facilitating the conversion of data into mental representations (Chen & Floridi, 2013). As a result, it can help people to see how the meaning of a words changes based on how they connect with other words over time.

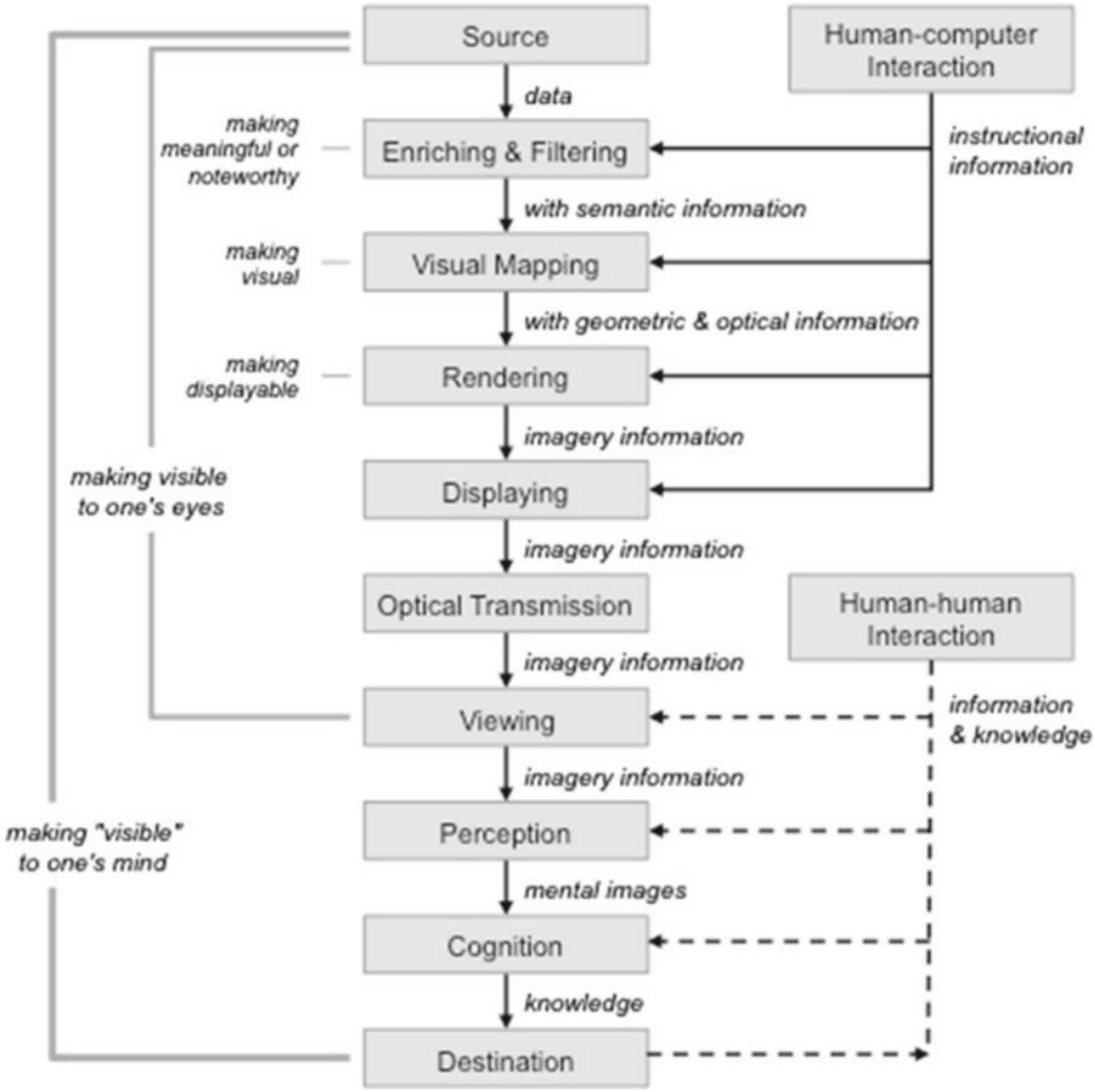

Figure 1: *A typical visualisation pipeline*

Source: *Figure 1 (Chen & Floridi, 2013)*

There are several visualisation techniques that have been proven to effectively capture and analyse semantic drift. These techniques include dimensionality reduction methods and network graphs (Tahmasebi, Borin, & Jatowt, 2021).

Dimensionality reduction techniques like t-distributed Stochastic Neighbour Embedding (t-SNE) plots (Tahmasebi, Borin, & Jatowt, 2021), offer a visual overview for observing how words such as 'woke' shift position in vector spaces over time, making shifts in meaning easier to detect. The use of t-SNE enables a word's semantic trajectory to be plotted over

time by reducing high dimensional vector data to a more interpretable 2D visualisation. These visualisations make abstract patterns of drift more interpretable by providing a more intuitive map of how the contextual environmental of word meaning can fluctuated over time.

However, interpretability is limited with this technique because vector-based methods such as embeddings are often unable to offer insight into how or why a semantic change has occurred (Dubossarsky, 2018). Therefore, semantic network graphs should additionally be used to trace relational word meaning change as they can provide a relational perspective between terms. They are often built from the similarity scores between terms across time slices, allowing for the meaning shift of a term to be tracked as it becomes increasingly connected to new terms and communities across a social media platform. This includes the visualisation of pejoration or appropriation that can occur online. For example, by comparing the node neighbourhood of 'woke' across different snapshots on a graph, we can inspect the relational drift of the word as it moves from social justice and activism discourse into a more politicised pejorative usage.

Dynamic relational graph diagrams are an example of how a semantic network visualisation graph can be used in Neo4j.

Time series graphs are also used to highlight key semantic change points to visualise when key shifts in meaning occur, especially around cultural or political events. These techniques can be enhanced by incorporating interactive elements that can zoom into the specific context of the change. Combining these techniques can support a visual methodology for tracing meaning change so that it is interpretable for use with the graph-based logic of Neo4j.

Neo4j provides a useful visualisation environment for exploring semantic drift using real-time graph traversals for a dynamic view of how word meaning or term association can change over time. Tools such as Neo4j Bloom enable visualisation that provides users with the capability to build and explore sematic network diagrams through interactive interfaces that support pattern recognition (Neo4j, 2022).

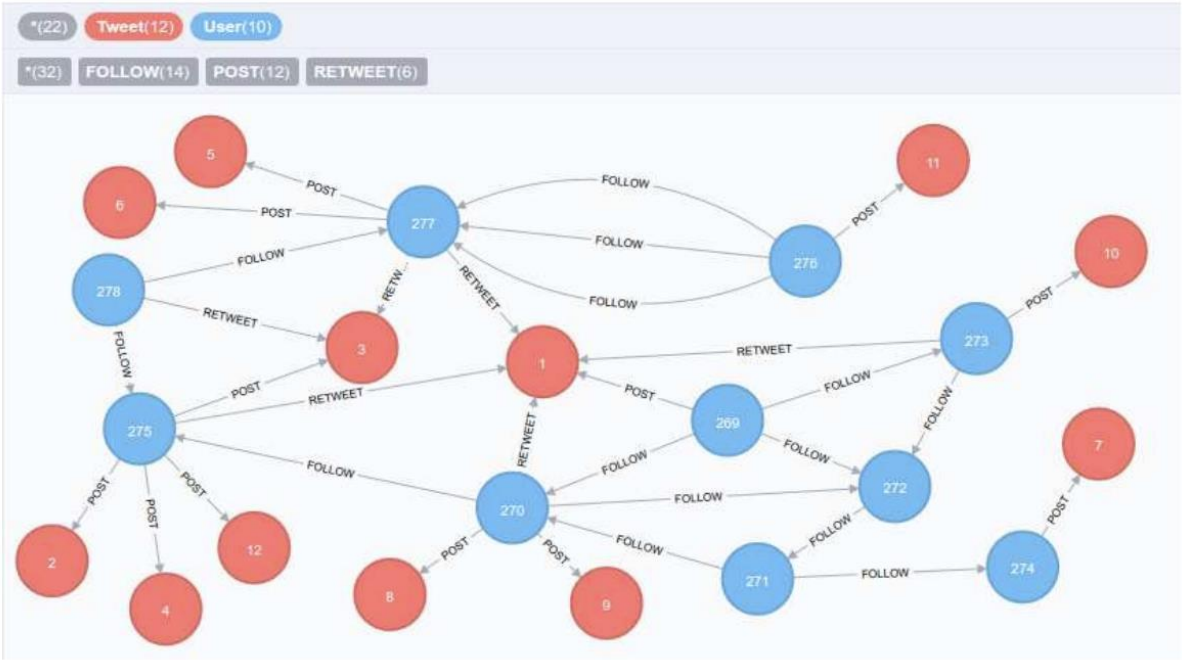

***Figure 2**: Social Media Database Design using Graph Database*

***Source:** Figure 1 (Aryono, 2016)*

Graph layouts that organise nodes according to the strength of their relationships, such as the model depicted in **Figure 2** that maps relational structures between users and tweets, allow patterns of semantic change to visually be observed.

Additionally, Neo4j Bloom has interactive features to support exploratory insight by allowing users to hover over node cards for more information, edit node properties for customisation, and stylise graphs through rule-base colouring (Neo4j, 2022). This makes Neo4j particularly well suited for capturing the instable dynamics of social media discourse through its ability to represent the evolving relationships between users and the contextual elements that shape the content that they interact with.

2.3.4 Summary and Evaluation

The study of semantic drift has shifted from a traditional theory-driven approach to a more computational data-driven method that is most suitable for processing large-scale data, such as the data produced from online media platforms.

Distributional modules have utilised these techniques to measure word meaning overtime in a scalable way within the data-driven method. However, these techniques are too reliant on

global word representation which do not consider the polysemy that exists for word meaning. Therefore, to overcome this we can incorporate time mapping techniques and change point detection (Tahmasebi, Borin, & Jatowt, 2021) to track how a word's contextual environment can evolve semantically across time slices. This uses the cosine distance that is mapped across vector embeddings to detect when and how a pejorative shift has occurred. Distributional semantics will be utilised as a computational approach to analyse the process of pejoration over time. This approach models changes in word meaning by examining how patterns of word co-occurrence vary in different contextual environments. Graph-based models can then be used to trace the semantic drift relationally.

Using Neo4j as a graph database, the semantic network graph can be visualised and queried. To expand upon the visualisation capabilities, Neo4j Bloom would provide a more expansive visualisation tool to support the interactive exploration of the word neighbourhood across time using interactive graphs and styling features. When combining these approaches together, it can provide a computationally robust framework for enabling the detection and visual interpretation of semantic drift.

3 Methodology

3.1 Research Hypothesis

The word ‘woke’ was hypothesised to have undergone pejorative semantic drift and was selected as the target word to observe for this investigation.

The central research hypotheses that guided this investigation can be seen below.

Research Question	Research Question
Q1	Are graph-based representations able to accurately illustrate the pejorative drift in the target word?
Q2	Are graph-based representations able to accurately illustrate the pejorative drift in the target word?

The main objectives for this investigation are given below.

Objectives	Research Objectives
1	To verify semantic drift of a word over time.
2	To quantify the pejorative declination of a word over time.
3	To assess the application of semantic analysis tools in identifying word pejoration.

***Table 4**: Main Research Objectives*

3.2 Research Design

This study adopted a computational sociolinguistic design to trace semantic drift in online discourse.

The pejorative meaning change observed in user-generated content on the social media platform YouTube was the central focus of this investigation.

To effectively capture the contextual and relational shifts in meaning over time, a hybrid approach that integrated graph-based modelling and distributional semantics was utilised to trace semantic drift in online discourse. This design allowed for the temporal mapping of context and co-occurrence patterns within YouTube video titles and user comments to reveal how word usage can fluctuate in response to digital interaction. It was based on the recognition that semantic drift, particularly within digital environments, is not only shaped by

the natural linguistic evolution of words, but also the norms that govern how users can interact with one another on online platforms. It was therefore important that both the structural and contextual aspects of meaning change were captured through this hybrid approach.

NLP techniques and network graph modelling were integrated under a sociolinguistic framework to analyse semantic drift. Graph-based visualisations in Neo4j depicted the evolving semantic neighbourhoods of the targeted term, while word embedding models and temporal analysis techniques quantified the trajectory of the pejoration. The fusion of both these methods was able to capture the structural shifts in word relationships within the network (e.g. the shifts in the relational proximity that occurs between words). Alongside the distributional patterns (e.g. the changes in the contextual similarity of a word neighbourhood over time). These tools enabled semantic drift to be modelled quantitavely to reveal how social patterns driven by community dynamics can drive how language used within these communities can change.

The research was built on insights from ethnomethodology and the study of linguistic variation across social groups. These theoretical foundations asserted that meaning is not fixed but reflexively constructed and negotiated through interactions that are shaped by situational contexts. Therefore, the design focused on how meaning can shift across time and change context between communities.

The design provided a methodological framework for investigating semantic drift as a computational process that is also a sociolinguistic phenomenon. It captured the influence of cultural discourse in online environments in shaping how meaning can become recontextualised and reshaped in a rapidly evolving communicative landscape.

3.2.1 Justification of YouTube API

According to the latest data from the Digital2024 Global Overview Report by Datareportal (DataReportal, 2024), YouTube is ranked as the world's most used social media platform in terms of active users on the app. With approximately 5.07 billion people, 62.6% of the world's global population, being identified as social media users.

I decided to use YouTube as it provided a rich linguistic dataset for analysing the semantic drift of user-generated data across different *community-driven* environments. YouTube was

selected due to its high user engagement and its distinct community to user interaction models.

YouTube provided an environment centred around language that is shaped around video commentary. By using data from this platform, it allowed this study to observe semantic variation in such a unique digital environment.

3.3 Data collection

The YouTube Data API v3 (Developers, n.d.) was utilised to extract time-stamped metadata relevant to the selected keyword of the investigation. The data collected from the YouTube API included video titles, video descriptions and top comments replies of the target word. A keyword-based search request, based on the target word 'woke', identified and retrieved up to 100 videos per year between the years 2010 to 2024.

Within python, I created a search function called `search_videos` **Appendix 1** to access the YouTube Data API v3 via the `googleapiclient.discovery` module in Python. The function does a keyword search for videos that contain the word 'woke' in their video title and or video description. Additional parameters were also added to the function to refine the scope of the search. These parameters included a limitation to search only videos and not playlist or channel names, as well as limiting the search to only English language videos.

An additional function called `fetch_top_level_comments` **Appendix 2** was created to get the "top-level comments" from the videos identified with the target word 'woke'. This was done by the function sending a request to the `commentThreads().list` under the `googleapiclient.discovery` module to extract 10 "top-level comments" in a plain-text format from the videos identified earlier based on their unique video ID.

The "top-level comments" are the replies that start the comment threads under a video. Using the "top-level comments" helped to focus the semantic analysis on texts that centred around the initial topic of discussion, reducing the semantic noise that could occur from deep reply-chains that may drift away into off-topic discussions. The "top-level" comments were collected under the variable `comment_text` by nesting the relevant dictionary keys, using the JSON response structure (Developers, CommentThreads, n.d.).

To enable year by year comparisons of how the target word was used online, the video metadata and user comments were collected separately for each year between 2010 and 2025. This allowed for separate analysis to trace the meaning change of the word, as well as the shifting contextual associations of the word overtime. Once retrieved, the video titles and their descriptions were merged into a single block of text to enable future natural language analysis. YouTube video titles can typically be short and punchy to catch the viewers' attention, whereas the description exists as a place for the Youtuber to offer a more elaborative explanation to the video's content. Therefore, I combined the two into a single text to provide a more complete context window to input into the NLP pipelines. Videos that did not have a video title and description or a video ID were excluded from the dataset to ensure interpretability in future analysis.

An error catching block was also added to identify any issues that may have occurred with extracting the data. As there are some API limitations, it helped to highlight areas where the limit had been reached.

All the raw metadata for each year was saved into its own CSV file to enable a modular evaluation of the semantic shift patterns year by year. **Table 5** is an example of some of the data saved for the year 2020

video_id	title	description	year
ht6z2d7riNg	The SHOCKING similarities between joining a cult and today's woke, social justice movements	"Woke" Americans, or those who describe themselves as Social Justice Warriors, should know they're actually members of a cult.	2020
O4mSqohd2Ow	Adam Coleman | What Does It Mean To Be Woke?	Adam Coleman's Channel https://www.youtube.com/user/scottlane7 Take the OMA Apologetics Crash Course ...	2020

Table 5*: Raw Metadata extraction from 2020 dataset*

3.4 Data processing and Tools

3.4.1 Preprocessing

All the video metadata and user-generated comments collected from YouTube Data API v3 underwent preprocessing to prepare the dataset for sentiment analysis. The preprocessing was implemented in Python using the NLTK modules for data cleaning and tokenisation.

Under the function `clean_text` **Appendix 3** each text was lowercased and removed of any punctuation, emojis and whitespace to ensure that it was standardised for use in the NLP sentiment analysis pipeline. The texts were further normalised using NLTK's `word_tokenize` function (NLTK Project), by filtering out common English stopwords (NLTK Project) such as 'the' and 'is' to reduce the noise to only meaningful data.

When performing sentiment analysis, it was important to remove the stopwords as they were not significant to the analysis and the construction of word embeddings. The tokens were then joined into a single string in preparation for vectorisation and natural language analysis. The cleaned and tokenised outputs for each year in the dataset ranging from 2010-2025 was stored separately to enable for later analysis of the year-by-year changes.

3.4.2 Sentiment Analysis

Sentiment Score Classification of Word Meaning

The NLTK library was used for sentiment analysis by applying a year-by-year VADER sentiment score (NLTK Project, n.d.) to each title, description and comment across the dataset. A VADER sentiment score is a compound score of the sum of all the word ratings in a text, normalised to a range of -1 to +1 to give an overall sentiment rating of the text (INFOpedia (University of Colorado Boulder)).

Each score was classified into three categories, positive 'pos', neutral 'neu', or negative 'neg' using the conventional VADER thresholds shown below:

- Positive: ≥ 0.05
- Negative: ≤ -0.05
- Neutral: between -0.05 and 0.05

The function `compound_score_classification` **Appendix 4** was then used to compute a compound polarity score that calculated and stored the average sentiment score per video aggregated by year. An overall sentiment label was also produced along with a distribution count of the number of videos that fell into each sentiment classification per year. This allowed for the temporal variation in sentiment of the target word to be traced across the years.

The resulting sentiment summary's yearly outputs were stored in a tabular format and exported to a CSV file for further analysis and visualisation. The information stored included the "Average sentiment score", "Overall sentiment label", and the counts of positive, neutral and negative videos.

Tracking the Distributional Shifts in Word Meaning using Word Embeddings

In line with **Section** 2.3.2 Methods for Detecting Semantic Drift**,** I developed a vector-based method for tracking semantic drift over time using word embeddings. This method utilised the spaCy `en_core_web_sm` model (spaCy (Explosion AI), n.d.) to capture the distributional changes in word meaning over time by extracting the contextual word vectors for all the instances of the target word across the tokenised YouTube texts year by year.
For every occurrence of 'woke' from 2010 to 2025, the vector representation was collected from spaCy's pipeline by iterating through the tokenised corpus. The vectors were then

averaged to compute a mean vector per year to represent the typical usage of 'woke' in that year.

```
doc = nlp("  ".join(sentence))
for token in doc:
if token.text.lower() == target_word:
vectors.append(token.vector)
```

Code Extract 1*: Token vectorisation*

Above is an extract for **Appendix 5** that shows the conversion of the token into a word embedding (vector) using spaCy. The mean vectors were stored under the variable `year_embeddings` to later be used for similarity analysis to compute the cosine distances between years.

The cosine similarity was computed between each year's mean vector and a baseline reference vector in order to measure and compare the degree of semantic drift across time. The baseline year was given as 2010.

```
base_vector = year_embeddings[base_year]

semantic_drift = {}

for year, vector in year_embeddings.items():
        similarity = cosine_similarity([base_vector],
        [vector])[0][0]
        cosine_distance= 1-similarity
        semantic_drift[year] = round(cosine_distance, 4)
```

Code Extract 2*: Computational Cosine-distance calculation*

Code Extract 2 is an extract from **Appendix 6**. This extract shows the computational representation of the **Equation 1**'s cosine-distance calculation, where similarity score was first calculated by importing the `cosine_similarity` from the `sklearn.metrics.pairwise` (scikit-learn developers) library and then deducted from 1 to

get the semantic drift score, `cosine_distance`. The closer the semantic drift score is to 1, the greater the word's meaning has shifted away from its original 2010 representation.

This method showcased semantic drift as a shift in distributional representation. The semantic drift scores were stored in an indexed dictionary and exported as a CSV for visualisation and analysis. The drift was then plotted to create a time series graph that showed the magnitude of semantic drift of the word 'woke' from its baseline meaning in 2010.

2010 was chosen as the reference point due to its positioning at the beginning of the dataset and its cultural relevance to it initialising the mid 2010s resurgence of social justice discourse highlighted by (Morris, 2020). Additionally, it was chosen due to the high average sentiment score that was associated with the word in that year. Therefore, using this as the anchor point provided a clear comparison for how the word's contextual associations had evolved over time.

To further illustrate how the semantic representation of 'woke' has evolved over time, a 2D visualisation was also created to project the yearly word embeddings using t-SNE to reduce the dimensionality of the vector space. This technique plotted the yearly average vectors given by the `year_embeddings` as arrowheads originating from the origin where each arrowhead is given a year label to indicate the trajectory of the drift for that year.

```
years = list(year_embeddings.keys())
vectors = np.array([year_embeddings[year] for year in years])

tsne = TSNE(n_components= 2, perplexity= 5, random_state= 42)
reduced = tsne.fit_transform(vectors)
```

***Code Extract 3**: t-SNE calculation*

Code Extract 3 is an extract showing the implementation of t-SNE, where `n_components` is the dimension of the embedded space and perplexity is the number of nearest neighbours to the target word used in the learning algorithm.

A perplexity of 5 was chosen due to YouTube video titles and descriptions typically being shorter as to quickly gain the user's attention into wanting to view and engage with the YouTube video.

3.5 Data Visualisation in Neo4j

3.5.1 Data preparation for Neo4j

Similar to the `fetch_top_level_comments` function created for compound sentiment analysis, the function `fetch_comments_metadata` was created to fetch the video titles and description metadata alongside the top-level comments for a given YouTube video, in preparation for the importation into the Neo4j graph database. The function parses through and packages relevant fields (e.g. comment text and user ID) into a dictionary format that is suitable for graph modelling in Neo4j.

It also utilises the SentimentIntensityAnalyzer module (NLTK Project) in the NLTK library to extract the compound polarity scores for each comment text.

YouTube allows for YouTube videos to have their comments disabled and are sometimes auto disabled for children's content or due to an enforcement of community guidelines. To ensure that videos that had their comments disabled did not stop the script, I added a filter to skip such videos.

The YouTube metadata and sentiment was then extracted from the YouTube API results retrieved earlier on during the data collection stage using the `fetch_comments_metadata` for each video. The data was saved into a CSV file called words.csv that contained the comment-level sentiment metadata across all years and videos that matched the target word 'woke'.

Subsequent CSV files were created in preparation to import into Neo4j for graph modelling. These included:

- words.csv: Comment-level metadata used to create Comment nodes that link the user with their comment and the comment to the unique video ID.
    - With fields: commentID, videoID, userID, commentTxt, year, token, title, description and sentiment.
- user.csv: Creating User nodes that link to an anonymised user ID.
    - With fields: userID and usernames.
- videos.csv: Video metadata used to create Video nodes for graph database querying.
    - With fields: commentID, videoID, userID and commentText

- context_word.csv: Stores co-occurrence patterns of the target word *woke* to construct a semantic neighbourhood for a temporal semantic drift network.
    - With fields: token, co_token, score and year

The co-occurrence patterns across the comments were extracted by tokenising each individual comment into word units using the NLTK `word_tokenize` function. To ensure uniformity across the tokens, the comment texts were lowercased with punctuation pairings being excluded to reduce noise and safeguard the quality of the co-occurrence graph.

The co-occurrence logic was implemented through the function `get_cooccurernce` **Appendix 7**. The function shown in **Code Extract 4** loops through each tokenised sentence to identify how frequently two words appear near to each other within the same sentence. I selected a `window_size` (word radius) of 10 to ensure that all possible co-occurrence pairings were identified.

```
    cooccurrences = Counter()
    for tokens in sentences:
        for i, token in enumerate(tokens):
            for j in range(i+1, min(i + window_size + 1,
len(tokens))):
                w1, w2 = tokens[i], tokens[j]
                if w1 == w2:
                    continue
                if w1 in string.punctuation or w2 in
string.punctuation:
                    continue
                if w1 in stop_words or w2 in stop_words:
                    continue

                pair = tuple(sorted([w1, w2]))
                cooccurrences[pair] += 1
    return cooccurrences
```

***Code Extract 4**: Co-occurrence function*

3.5.2 Graph construction in Neo4j

The CSV files words.csv, user.csv, video.csv and context_word.csv were imported into the Neo4j database in order to build a graph network for semantic drift analysis.

I first mapped some constraints to the databases to prevent the duplication of nodes and to enforce uniqueness for the key node identifiers. These constraints were created using CREATE CONSTRAINTS… IF NOT EXISTS structure, as it helped to maintain the semantic clarity of the dynamic graph produced in Neo4j. This was especially true for the word co-occurrence properties of the Word nodes as they required multi-property constraints. The 4 constraints detailed below were individually run in Neo4j.

```
CREATE CONSTRAINT word_unique IF NOT EXISTS
FOR (w:Word) REQUIRE (w.token, w.year, w.commentID) IS NODE KEY;
```

```
CREATE CONSTRAINT user_unique IF NOT EXISTS
FOR (u:User) REQUIRE u.userID IS UNIQUE;
```

```
CREATE CONSTRAINT video_unique IF NOT EXISTS
FOR (v:Video) REQUIRE v.videoID IS UNIQUE;
```

```
CREATE CONSTRAINT video_id_unique IF NOT EXISTS
FOR (v:Video)
REQUIRE v.vidoeID IS UNIQUE;
```

An initial graph was then created using 6 Cypher ‘LOAD CSV’ queries.
The first 3 LOAD CSV cyphers were to create Nodes for ‘Word’, User’ and ‘Video’.

```
LOAD CSV WITH HEADERS FROM 'file:///words.csv' AS row
CREATE (:Word {
    token: row.token,
    year: toInteger(row.year),
    sentiment: toFloat(row.sentiment),
    commentID: row.commentID,
    userID: row.userID,
    videoID: row.videoID
});
```

This cypher created Word nodes to represent the target word ‘woke’ that is under investigation. This node is the foundation of the semantic graph. It represents each instance the target word in its year by year context.

```
LOAD CSV WITH HEADERS FROM 'file:///users.csv' AS row
CREATE (:User {
    userID: row.userID,
    username:row.username
});
```

This cypher created User nodes linked to the unique user Ids and usernames. The Users were modelled to capture the types of different communities that would shape semantic change.

```
LOAD CSV WITH HEADERS FROM 'file:///videos.csv' AS row
CREATE (:Video {
    videoID: row.videoID,
    title: row.title,
    description: row.description,
    uploadYear: toInteger(row.uploadYear)
});
```

This cypher created or merged Video node using their unique video Id to store YouTube video metadata.

I then created relationships between the_.

The first relationship created was between userID and commentID. The cypher run in the Neo4j browser created 6045 relationships.

```
LOAD CSV WITH HEADERS FROM 'file:///words.csv' AS row
MATCH (u:User {userID:row.userID}), (w:Word {commentID:
row.commentID})
CREATE (u)-[:USED]->(w);
```

This query created `USED` relationships between userID and commentID to create 6045 relationships. It linked each User node to the Word node to identify the users that used the target word on the platform.

```
LOAD CSV WITH HEADERS FROM 'file:///words.csv' AS row
MATCH (w:Word {commentID:row.commentID}), (v:Video {videoID:
row.videoID})
CREATE (w)-[:MENTIONED_IN]->(v);
```

`MENTIONED_IN` relationships were created by connecting each Word node to the Video node for where it was used. This relationship was created to support the analysis how words can shift in meaning depending on the context in which they are used.

```
LOAD CSV WITH HEADERS FROM 'file:///context_words.csv' As
AS row
WITH row, toInteger(row.year) AS year
MATCH (w1:Word {token: row.token, year: year})
MATCH (w2:Word {token: row.co_token, year: year})
MERGE (w1)-[r:CO_OCCURS_WITH]-(w2)
SET r.score = toFloat(row.score);
```

This query created co-occurrence `CO_OCCURS_WITH` relationships to establish a link between the words that appear together around the target word per year with a co-occurrence score. This co-occurrence score is a weighted numerical value that represents strongly two words appear together. The greater the co-occurrence score, the more frequent the words occur together. This would suggest a strong contextual connection supporting analysis on semantic drift.

Visualisation of schema

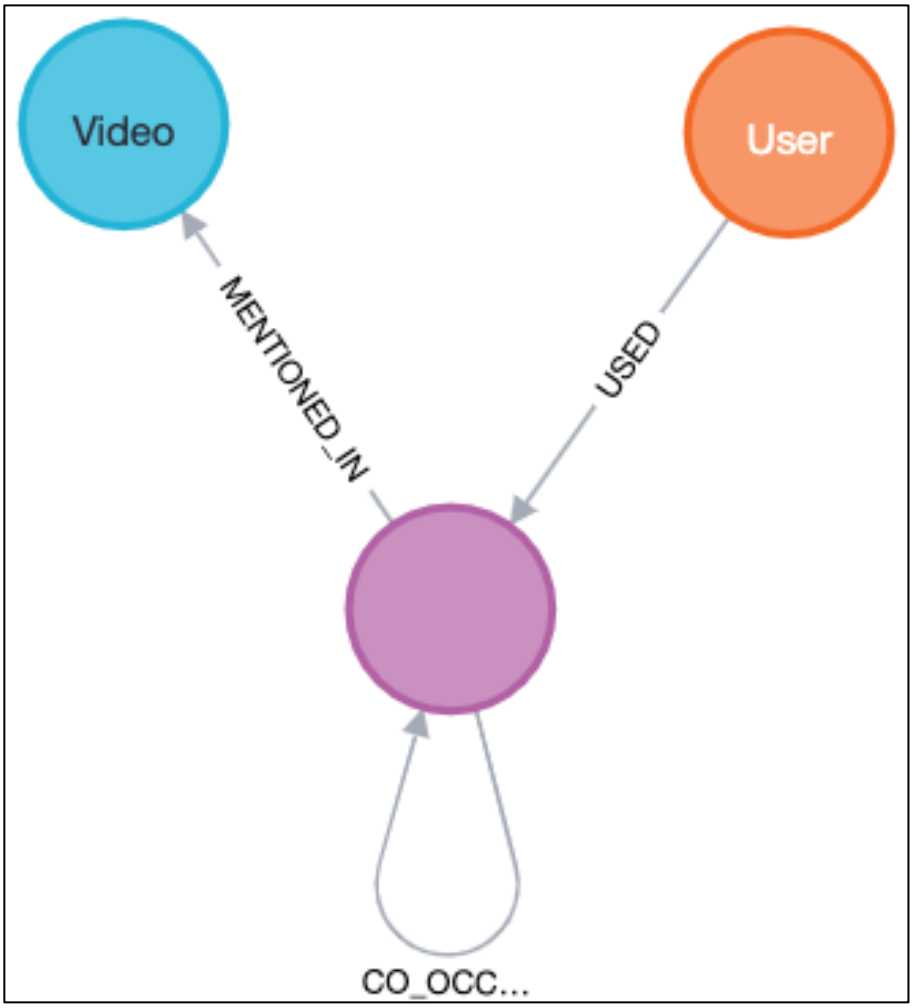

***Figure 3**: YouTube Database Schema Visualisation*

Figure 3 is the visualisation of the schema structure that was produced of my dataset. On the Figure, the purple circle represents the Word node, the blue circle the Video node and the orange circle the User node. The Word node was the main token and comment that structures the foundation of all the relationships. The Video node represented the video metadata, and the User node represented the user who posted the comment.

Figure 3 shows a direction relationship from User to Word (`USED`) and from Word to Video (`MENTIONED_IN`). The `CO_OCCURS_WITH` relationship with is represented as an undirected relationship between the two Word nodes, this would be the co-occurrence pairs.

3.6 Ethics

All the data used in this study was publicly available and did not require direct contact with any human subjects.

To ensure alignment with YouTube's Terms of Service and ethical research protocols, only data from public videos were retrieved.
Where users had removed or privatised their video content or account, the data was then removed from the dataset.

To ensure user privacy and data integrity, all retrieve data was:

- Pre-processed to remove any identifying metadata.
- Stored in an anonymized format for data analysis including the anonymisation of usernames.

The study focused on the semantic change patterns of language used in digital discourse. Therefore, it was not necessary to identify any individual patterns that may come form user behaviour. Insights into induvial user behaviour on the digital platform was not performed, in order to ensure user privacy and compliance with the ethical data handling standards.

4 Results

4.1 Sentiment Analysis

4.1.1 Semantic Scoring

Year	Average sentiment score	Overall sentiment label	No. Postive videos	No. Neutral videos	No. Negative videos
2010	0.1891	pos	169	141	67
2011	0.2012	pos	191	216	57
2012	0.2317	pos	165	107	57
2013	0.2173	pos	146	167	29
2014	0.265	pos	121	93	23
2015	0.1729	pos	149	170	59
2016	0.1162	pos	145	187	68
2017	0.1653	pos	182	159	78
2018	0.0966	pos	136	184	78
2019	0.1313	pos	159	162	82
2020	0.0515	pos	184	133	145
2021	0.0419	neu	170	186	149
2022	0.0307	neu	183	174	168
2023	0.0028	neu	206	138	183
2024	0.0285	neu	185	188	156
2025	-0.0261	neu	188	151	211

***Table 6**: YouTube Sentiment Summary Score per Year CSV*

Table 6 presents the average semantic score of the word ‘woke’ per year as well as the distributional counts for the number of positive, neutral and negative videos for each year.

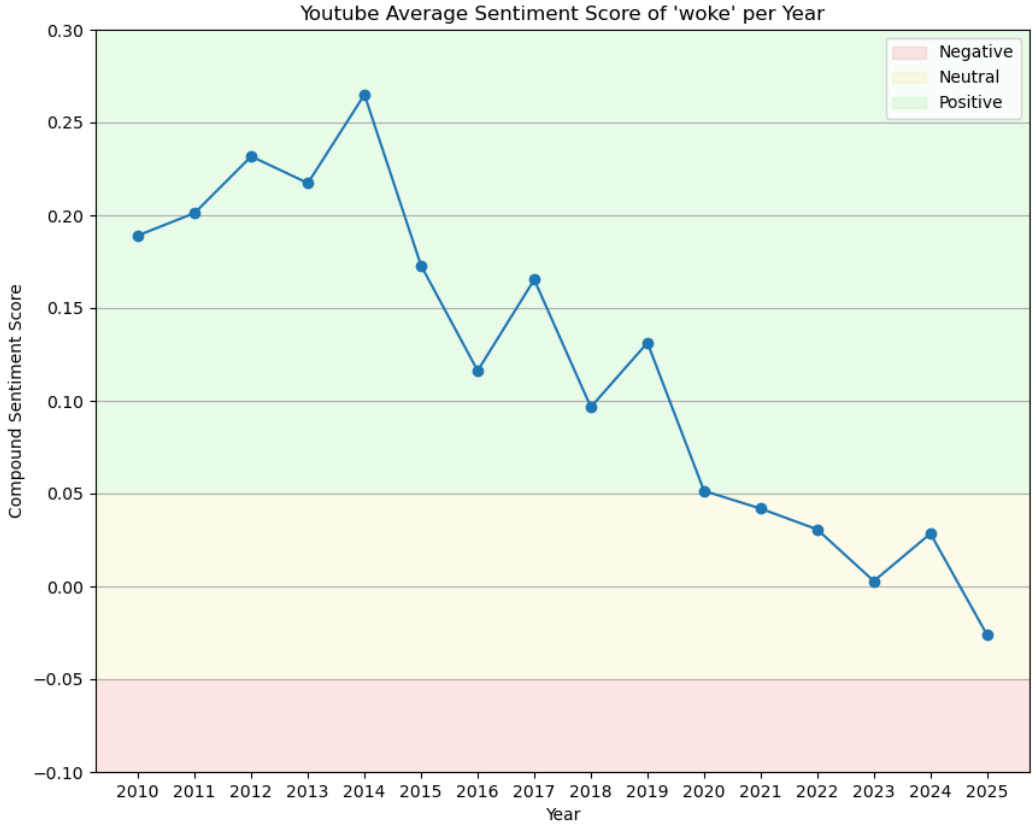

***Figure 4**: YouTube Sentiment Summary Score per Year plot*

Figure 4 is a semantic score line graph that depicts the compound semantic score of the word 'woke' from 2010 to 20225.

Figure 4 demonstrates a clear downwards trend in the compound sentiment score of the word woke' from the years 2010 to 2025. This pattern supports the claims by (Morris, 2020) that woke has undergone a pejorative transformation of meaning.

The figure offers a numerical metric for tracking pejoration, quantifying the pejorative declination of the word 'woke'. Its presentation of the target word's declination in sentiment shows a successful visualisation of word sentiment over time.

From years 2010 to 2014, we see an overwhelmingly positive sentiment in the use of the word 'woke'. With a slight incline in score from 0.1891 to 0.2650 from 2010 to 2014, comfortably nesting within the 'positive' sentiment score classification. However, between the years 2014 to 2025, we observe a gradual transition of sentiment where the sentiment

score for ‘woke’ eventually leaves the ‘positive’ categorisation to enter ‘neutral’ territory from 2021 onwards.

Table 6 shows the number of negative videos gradually increasing to eventually supersede the number the positive videos by 2025. The ratio of positive to negative videos changes from 169:67 to 188:211 from 2010 to 2025. This numerical shift provides quantitative evidence to the pejorative trajectory that ‘woke’ was theorised to have undergone in **Section 2.2.3 The Sociopolitical Dimension of Drift.**

Section 2.1.2 Types of Semantic Drift highlights how semantic drift is characterised by the shifts in the word usage, therefore causing the divergence of its usage away from its original meaning. The results given in **Figure 4** clearly show how this drift can be observable over time. It also provides empirical grounding for identifying the moments in time where this drift has occurred, with the ability to link it to the real-world sociopolitical discourse that may be causing the drift.

(Morris, 2020) outlines that ‘woke’ has undergone a weaponised reinterpretation as it gained visibility after the 2010 ‘Black Lives Matter’ movement and was recontextualization to a more unforgiving community. This process can be seen being captured in **Figure 4** by the declining sentiment score and in **Table 6** by the increase of negative videos as the years progress.

We do not see the line plot of **Figure 4** entering the category of ‘negative’ sentiment. However, this may be caused by the sentiment analysis not fully identifying the sarcastic or ironic uses of ‘woke’ as discussed in **Section 2.2.3 The Sociopolitical Dimension of Drift**.

4.1.2 Distributional Representations of Semantic Drift using Word Embeddings

Year	Semantic Drift from 2010
2010	0
2011	0.3548
2012	0.084
2015	0.1096
2016	0.1251
2017	0.128
2018	0.0791
2019	0.11
2020	0.0962
2021	0.0652
2022	0.0714
2023	0.0619
2024	0.0957
2025	0.0836

***Table** 7: Cosine distance calculations*

Figure 5 is a cosine similarity graph that plots semantic drift computed as cosine distance over time. The resulting plot illustrates an upwards trend in cosine distance, reflecting a divergence of target word's semantic representation away from its original usage.

From the corresponding **Table 7**, we can observe a sharp increase in cosine distance in the year 2011 to a drift score of 0.3548, indicating the first strong divergence of the semantic meaning of 'woke'. This rapid shift early in the plot potentially marks the moment in which the word 'woke' had begun to acquire new sociopolitical meanings within online discourse.

From 2012 to 2020, the semantic drift fluctuates within the values 0.07 to 0.02. The semantic drift then stabilises at a lower level and persistent level from 2020 onwards, resting between 0.05 to 0.1. The stabilisation of semantic drift in these later periods suggest that 'woke' has undergone significant reconfiguration and evolved in more constrained and restrictive semantic environments that can be associated with its pejorative appropriation.

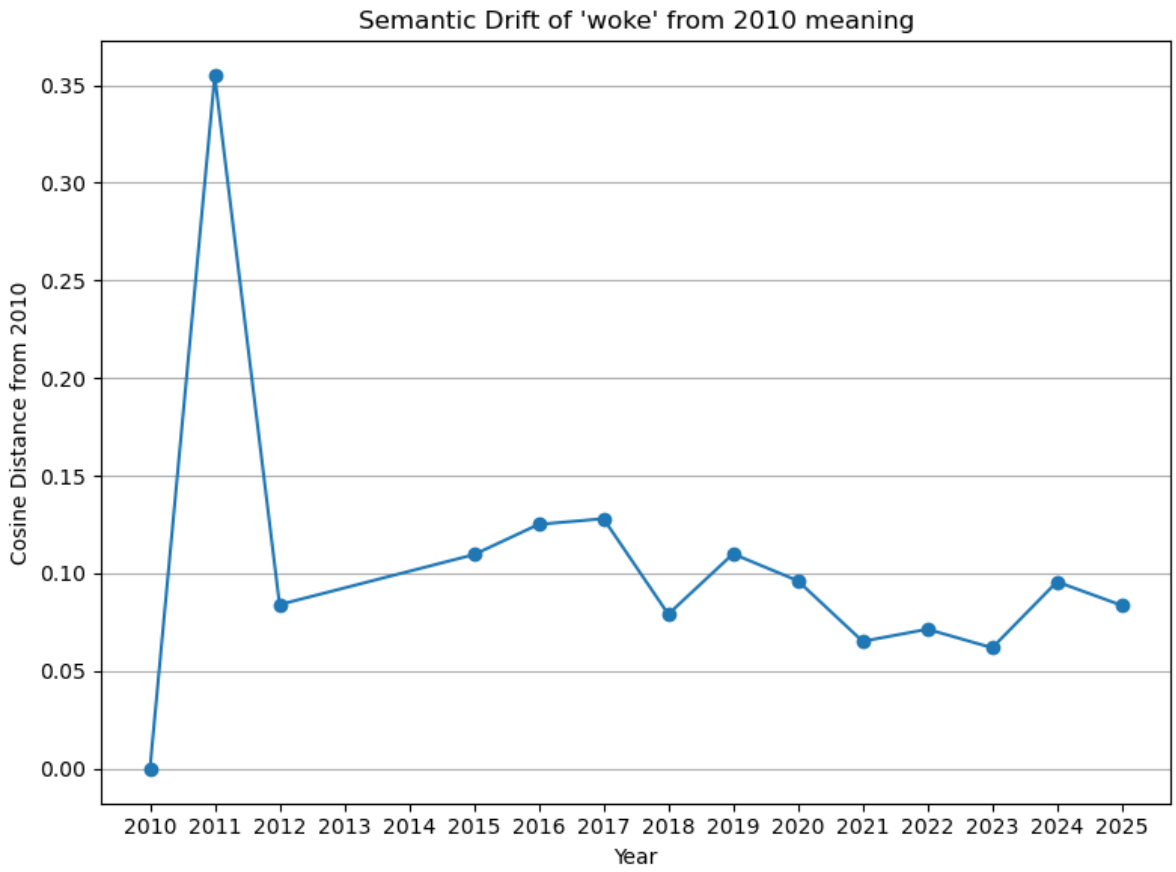

Figure 5*: Cosine similarity plot*

Most importantly, **Table 7** shows us that no subsequent years has a cosine distance return to near 0, confirming that 'woke' has undergone semantic change and has not retained its initial positive usage in social justice and awareness discourse as discussed in (Morris, 2020) from earlier years.

Unlike **Figure 4**, **Figure 5** does not prove semantic pejoration of the word 'woke', but rather it supports the view that 'woke' has experienced rapid semantic transformation to where the meaning has now settled into a more ironic usage as described in **2.2.3 The Sociopolitical Dimension of Drift**.

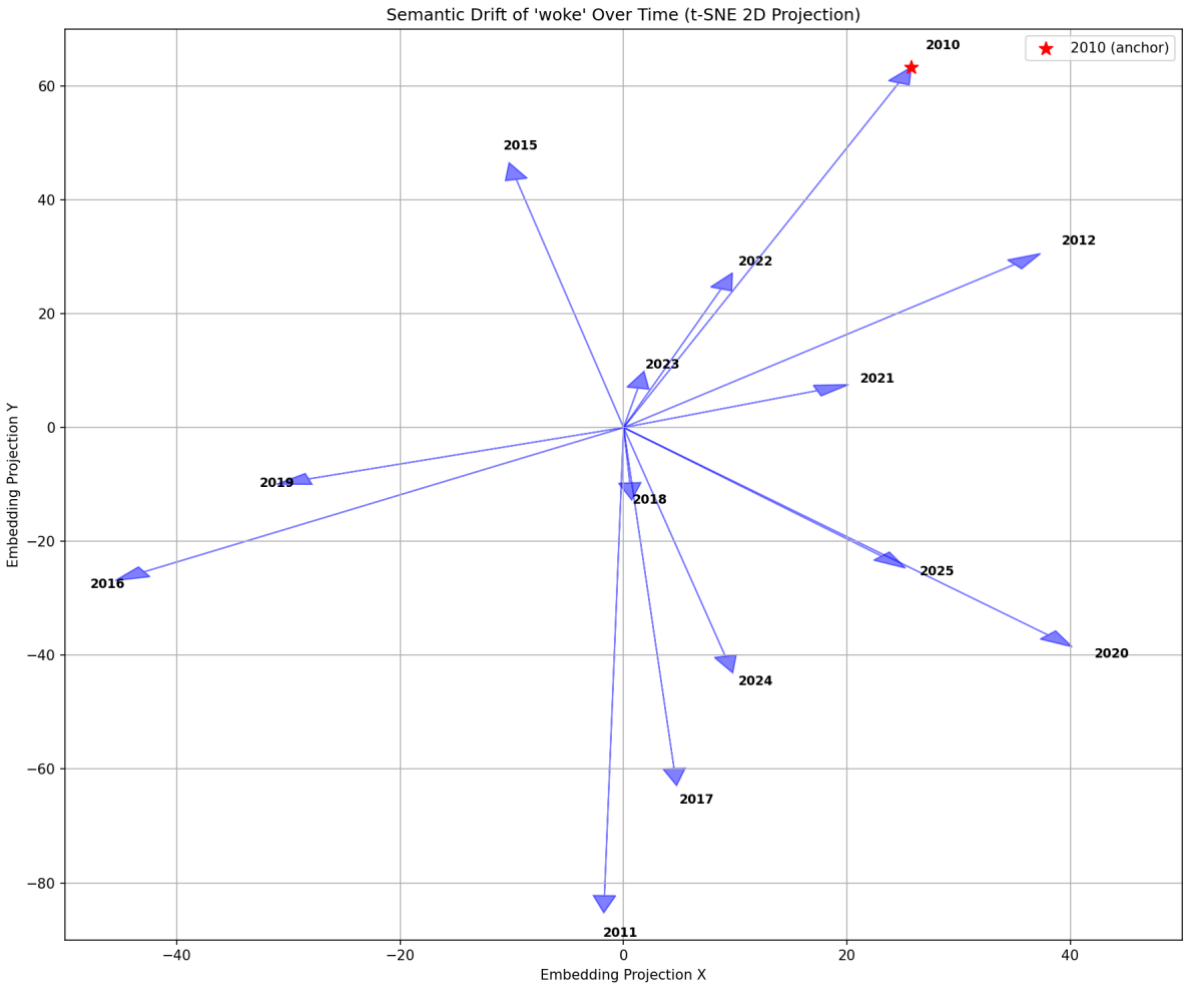

***Figure 6**: t-SNE 2D Projection of Semantic Drift*

Figure 6 builds on the approaches on representing distributional shifts in semantic drift by employing a t-SNE to reduce the high dimensional spaCy word embeddings into an interpretable 2D vector space.

There is variation in the directions that the arrowheads are pointing in, revealing significant divergences of meaning from the 2010 baseline (anchor). The red star represents the anchor year for which the following years trajectories are compared to.

The broad spatial dispersion of the arrows shown in **Figure 6** reflects a non-uniform evolution of meaning. This supports the claims made in **Section 2.1.5 Contextualising Drift in Digital Language Use** that emphasises the role of context in shaping the meaning of a word as indexical and socially negotiated. In this case, the evolving online political discourse is shaping how ‘woke’ is encouraged to be used within video content online.

The most significant transformation is shown to have occurred in 2011, indicating the start of increased discourse around the word ‘woke’. Where the arrowheads are seen to be tightly clustered, it suggests a period of meaning stabilisation, as also showcased in **Figure 5**.

Figure 6 reveals these periods most prominently to be from the years 2018-2023, where the representation of woke appears to settle withing a more uniform semantic environment.

4.2 Cypher Query Insights from Neo4j Graph Database

After creating my Neo4j database, multiple queries where ran to anslyse semantic drift overt time.

```
MATCH (w:Word) RETURN count(w);
```

The database was found to have 6062 unique Word nodes. Each of these nodes represents a token that was extracted from the YouTube comment. Each token represents a word. Therefore, the database contains a vocabulary of over 6000 words.

```
MATCH ()-[r:CO_OCCURS_WITH]->() RETURN count(r);
```

There are currently a total of 2944 co-occurrence relationships in the database.

For the years 2017 and 2020, I was able to retrieve the top 3 words that occurred with the target word ‘woke’. The data was by calling the query below, interchanging the year where necessary.

```
MATCH (:Word {token: 'woke', year:2017})-[r:CO_OCCURS_WITH]->(w2)
RETURN w2.token, w2.year, r.score LIMIT 3;
```

The top 3 word that co-occurred with the word ‘woke’ in 2017 was “stay”, with a co-occurrence store of 13.0.

In 2020, the top 3 words that co-occurred with the word ‘woke’ was:

1. 2020 (score = 3.0)
2. radio (score = 3.0)
3. station (score = 1.0)
4. call (score = 1.0)
5. letters (score = 1.0)

The presence of the word “stay” occupying the top 3 spots for the strongest co-occurring token to ‘woke’ in 2017 suggests strongly of a positive sentiment towards social justice discourse during that year. Within the mid-2010s Black Lives Matter resurgence, the phrase “stay woke” gained popularity online and was used as a call to raise awareness to racial injustice. The high co-occurrence of “stay” indicates the possible frequent use of the phrase “stay woke”, highlighting how the term ‘woke’ was still being used within its original sociopolitical context.

However, the top co-occurring words in 2020 suggest a shift away from its *call-to-action* usage towards broader discourse about how ‘woke’ is perceived in society. The top 3 words being “2020”, “radio” and “station” aligns with the idea that ‘woke’ has begun to undergo pejoration in the sense of mocking or as part of polarised political rhetoric online.

```
MATCH (:Word {token: 'woke', year:2017})-[:CO_OCCURS_WITH]-
>(w2017)

WITH collect(w2017.token) AS tokens_2017

MATCH (:Word {token: 'woke', year:2020})-[r:CO_OCCURS_WITH]-
>(w2020)

WHERE NOT w2020.token IN tokens_2017

RETURN DISTINCT w2020.token, r.score

ORDER BY r.score DESC;
```

This query returned words that co-occurred with ‘woke’ in 2020 but did not appear in 2017. The some of the words that were returned were “2020”, “sounds”, “radio” and “station”. This data shows that ta new contextual neighbour was created for the word ‘woke’ after 2017.

The top token returned was “2020”. It is most likely that that token refers to the year ‘2020. This is because, during this period many sociopolitical events were occurring including BLM protests, the US election and COVID-19. The score 3.0 for the token could indicate a high usage of the term ‘woke’ during these major events.

```
MATCH (w:Word {token: 'woke'}) RETURN w.year, avg(w.sentiment) AS
avgSent ORDER BY w.year;
```

The average sentiment score of the YouTube comments containing the word ‘woke’ was also retrieved through querying. The results can be seen in **Table 8**, where w.year in the year that ‘woke’ appeared in the comments and avgSent is the average sentiment score of the comments for each year.

From 2010, we see a high sentiment score of 0.4131 (rounded to 4 significant figures), showing a strong positive used of the word ‘woke’ during that period. Also possible suggesting that the word is still being used in its original context. However, as the years progress, average sentiment begins to decrease.

At first the decrease is rapid, suggesting heavy criticism or polarising discourse causing the contesting of the meaning. The decrease then gradually stabilises signifying more mixed usage of the term that could include sarcastic or ironic tones that are difficult to computationally differentiate.

w.year	avgSent
2010	0.4131125
2011	0.20478372093023256
2012	0.3503839285714284
2013	0.2617746987951807
2014	0.27757209302325586
2015	0.3050909090909091
2016	0.2368094117647059
2017	0.2205985294117647
2018	0.16655129870129867
2019	0.1525582352941177
2020	0.21646826923076928
2021	0.18338798882681576
2022	0.09536019
2023	0.12594072580645174
2024	0.06702531
2025	0.05901934

Table 8: *Average Sentiment Score of YouTube comments from Database*

```
MATCH (w:Word {token: 'woke'})-[r:CO_OCCURS_WITH]-(n)
RETURN w, r, n
LIMIT 10;
```

To visualise the local co-occurrence network of the word ‘woke’, I called a cypher query to return or the co-occurrence relationships across all the years. The co-occurrence subgraph produced illustrates the immediate semantic context of the word ‘woke’.

```
MATCH (w:Word {token: 'woke', year:2020})-[r:CO_OCCURS_WITH]-(n)
RETURN w, r, n
LIMIT 100;
```

The query above also produces a full co-occurrence subgraph but only for the co-occurrence relationships for year 2020 year. The co-occurrence graph produced revealed a densely connected network of terms that were used alongside ‘woke’ for that year.

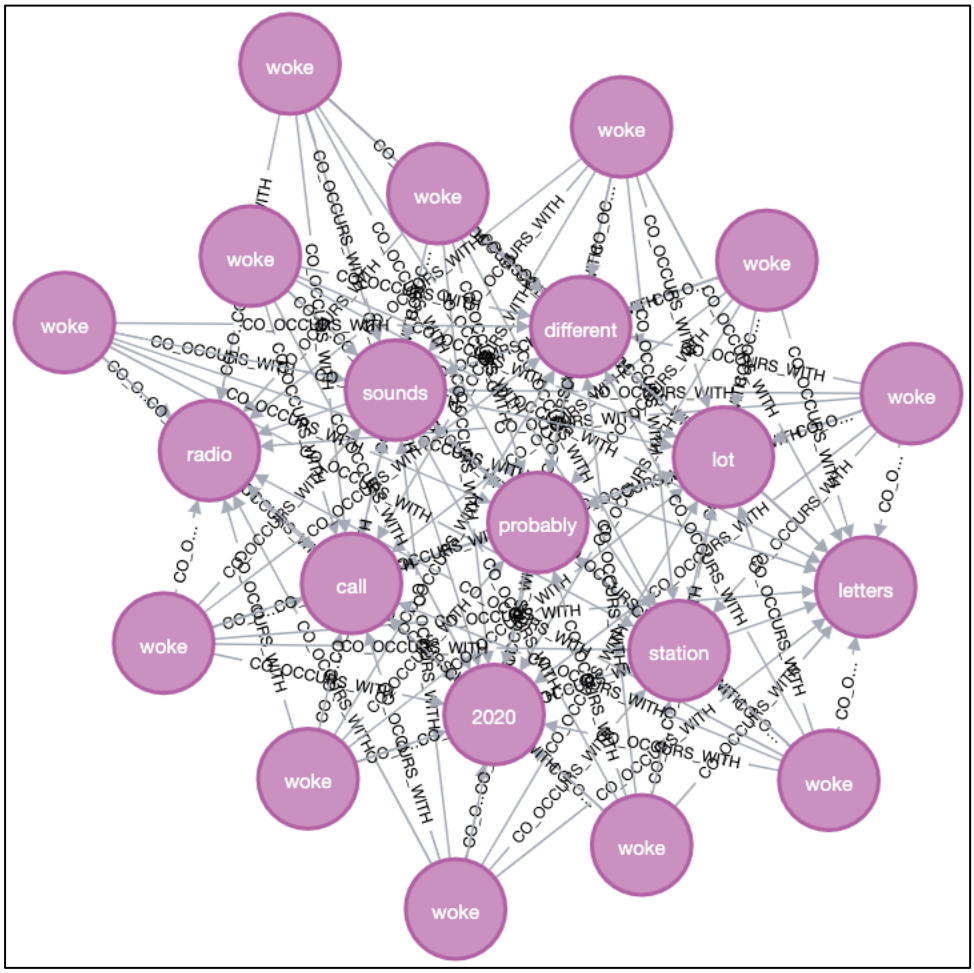

***Figure** 7: Co-occurrence network of 'woke' in 2020 with token*

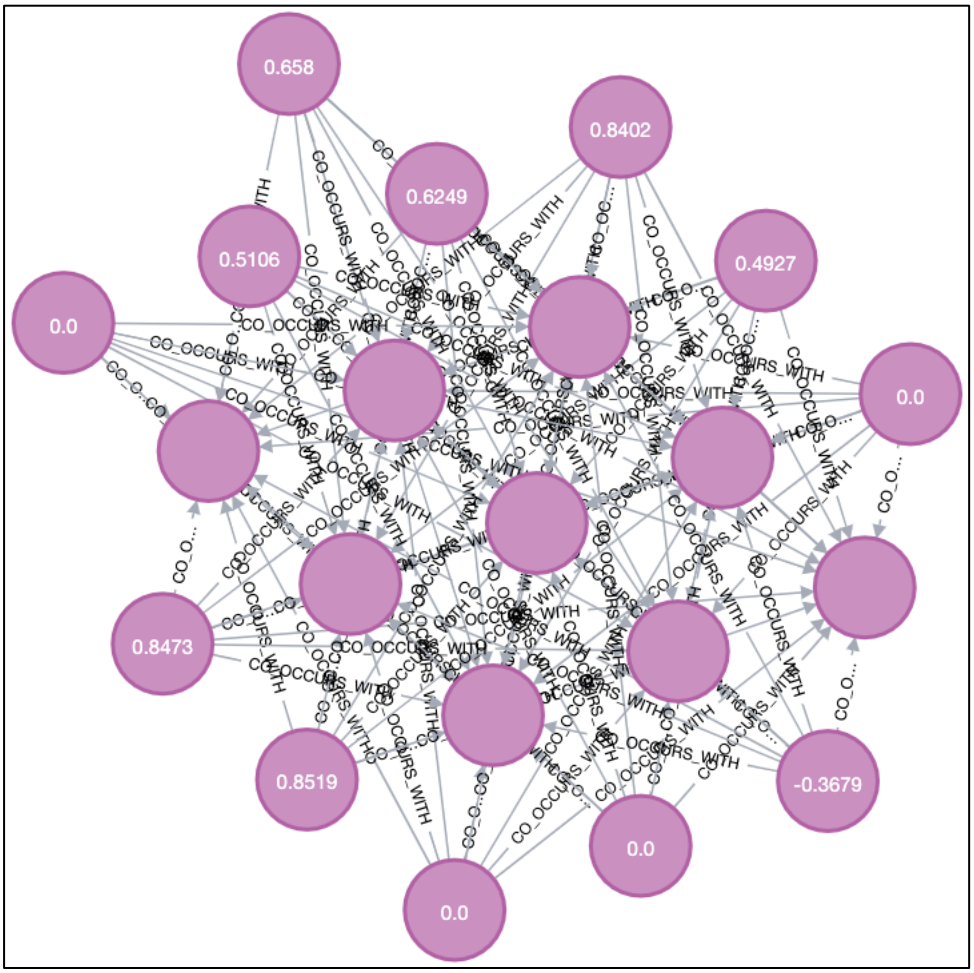

Figure 8*: Co-occurrence network of 'woke' in 2020 with sentiment score*

The sentiment scores given in **Figure 8** suggest a mixed sentiment environment with some strong positive sentiment scores of 0.8473, 0.8402 as well as some negative and neutral scores (e.g. -0.3679 and 0.0). The presence of negative and neutral scores supports the claim that 'woke' had undergone a pejorative shift.

Unlike the earlier years where the high sentiment of the target word was because of its co-occurrence with the word "stay", likely from the phrase "stay woke". We see a broader variety of co-occurrence showing how the meaning of 'woke' has diffused within online discourse. This is shift away from its original usage showing that the network is able to illustrate ongoing semantic drift.

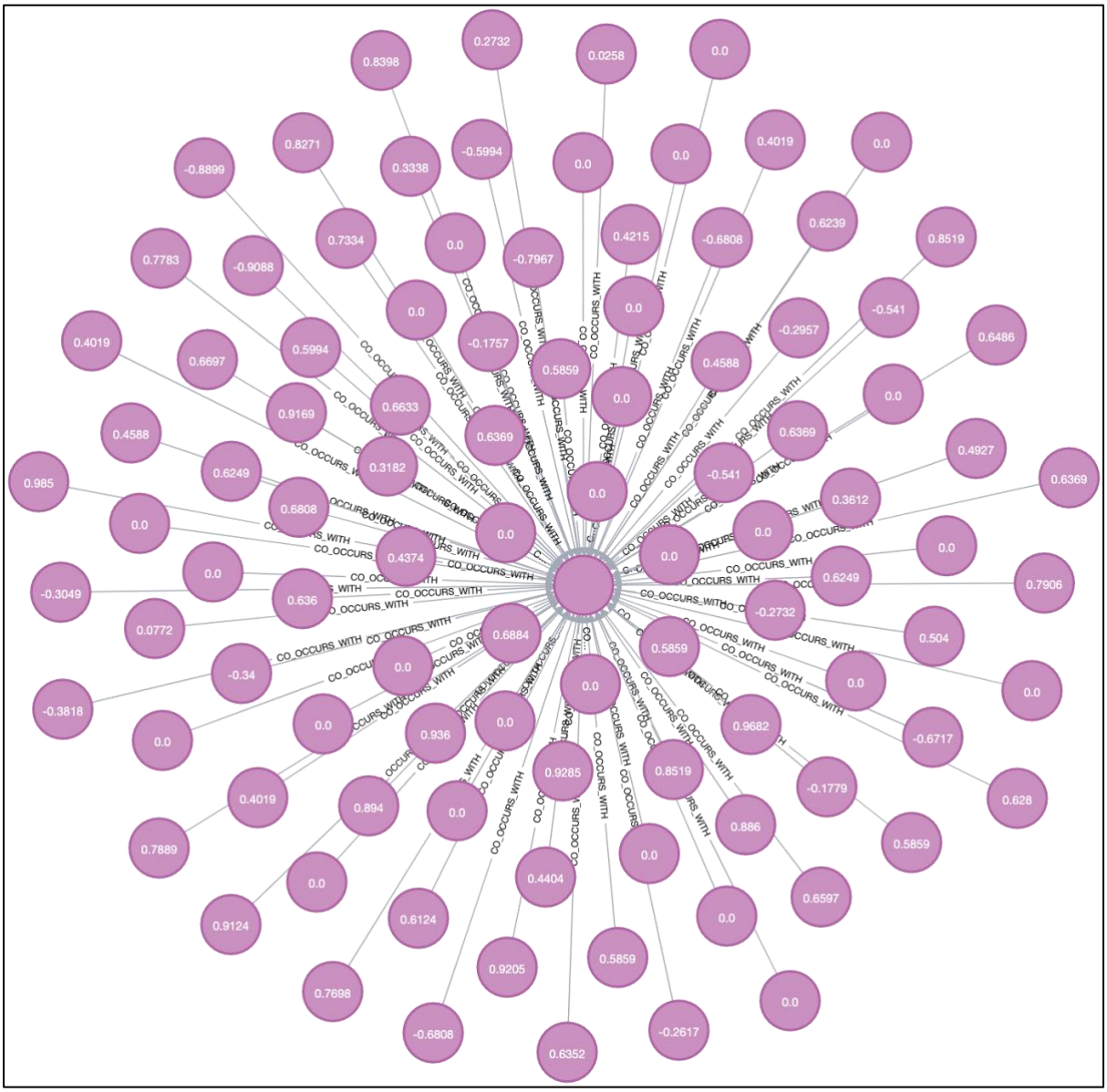

Figure 9:*Co-occurrence network of 'woke' in 2017 with sentiment score*

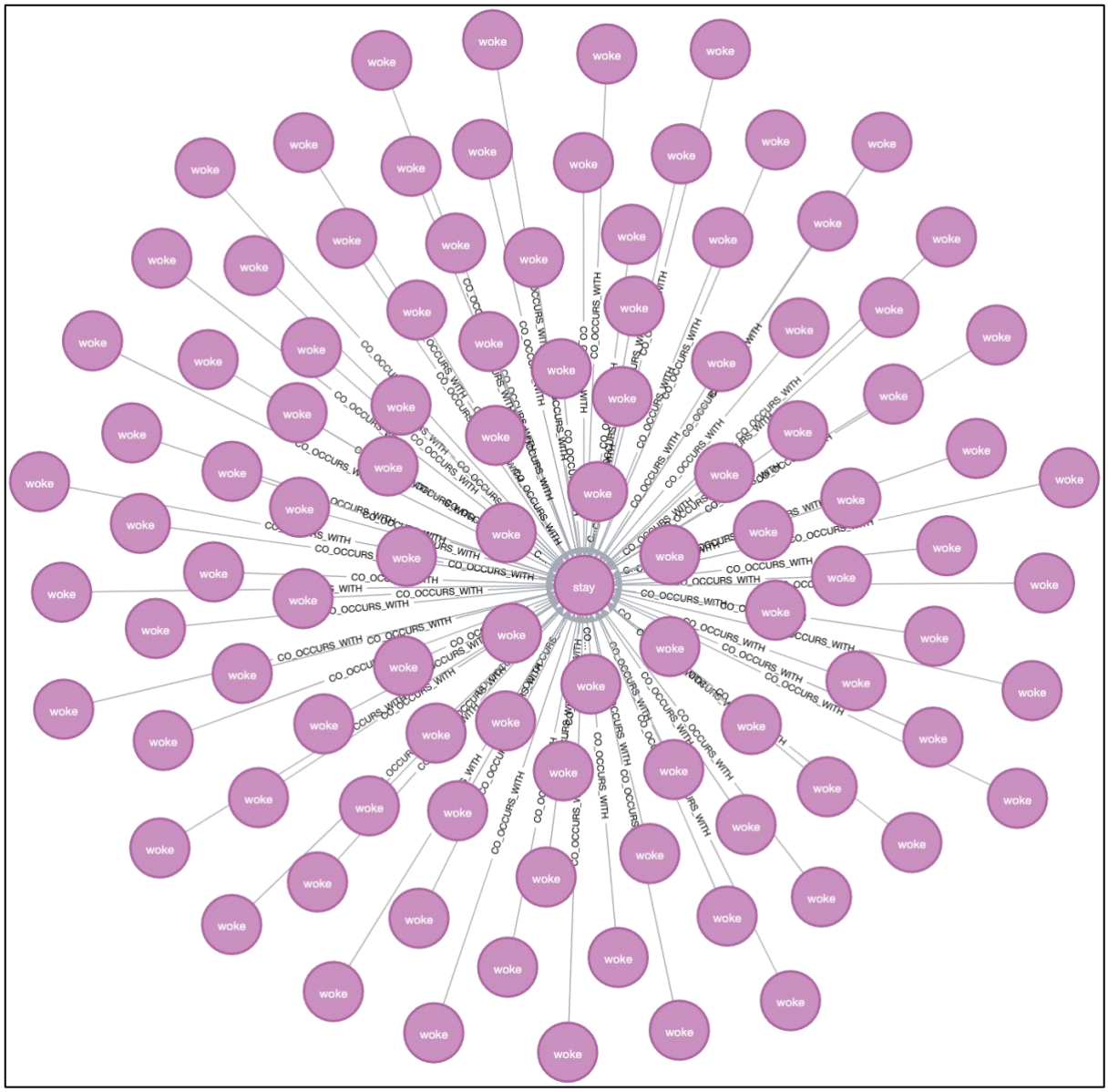

Figure 10:*Co-occurrence network of 'woke' in 2017 with token*

4.3 Discussion

Below are the objective given earlier in the dissertation.

Objectives	Research Objectives
1	To verify semantic drift of a word over time.
2	To quantify the pejorative declination of a word over time.
3	To assess the application of semantic analysis tools in identifying word pejoration.

Objective 1:

Objective 1 was focused on being able to identify semantic drift. Through this investigation, semantic drift was successfully able to be identified through the computational approaches detailed in **Section 2.3.2 Methods for Detecting Semantic Drift.**

By conducting a year-by-year analysis of the YouTube metadata focused of video title, video description and the top-level comments that contained the target word 'woke', a dataset was able to be complied for sentiment and distributional analysis. That dataset spanned from the year 2010 to 2025. The word 'woke' was identified to have undergone a pejorative semantic shift where co-occurrence pattern also identified a changing contextual window with the word over time.

Objective 2:

This objective was able to be met by applying a hybrid methodology that combined VADER sentiment analysis with word embeddings to provide a quantifiable metric for how a words meaning has changed.

The VADER model computed compound sentiment scores for each comment and video, revealing a declining trend of sentiment polarity from 2010 2025. Word embeddings were used to calculate cosine distances to measure the semantic shift of 'woke' over time, using 2010 as a baseline reference.

This approach, paired with the t-SNE visualisation further illustrated how the word ‘woke’ experienced a shift in its contextual neighbourhood. These techniques together provided measurable evidence of pejoration that can occur in online discourse.

Objective 3:

This objective was achieved by the creation of a graph-based model constructed in Neo4j. The model established Word, User and Video nodes linked to one another by the relationships `USED`, `MENTIONED_IN` and `CO_OCCURS_WITH`. These relationships were able to construct semantic neighbourhoods that can be analysed to see how a word's meaning can change over time.

The visualisation generated in Neo4j had weighted co-occurrence scores for each relationship so that the contextual environment of the word can be examined. The visualisation was able to reveal some structural shifts of the word ‘woke’ sentiment, supporting the interpretation that meaning change has occurred.

5 Evaluation

This study was successfully able to demonstrate that semantic drift can be computationally detected and visualised using a combination of sentiment analysis, word embeddings and graph-based modelling.
For all the processes applied within the overarching sentiment analysis investigation, the target word ‘woke’ demonstrated a clear declination of sentiment overtime with significant shifts in its distributional and representation. This supporting the hypothesis drawn from the social linguistic theory discussed earlier within this dissertation. The Neo4j graph database provided a further relational interpretation of the semantic shift that can occur for the word ‘woke’ by capturing the evolving co-occurrence patterns.

However, during the investigation some challenges occurred that caused some limitations to the scope of the analysis. During the creation of the Neo4j graph database, the network visualisation of the co-occurrence relationships was only able to successfully be generated for the years 2017 and 2020. Due to this limitation, the continuous semantic drift of the word that

occurs within the full-time frame of 2010 to 2020 (the scope of this study) could not be observed, therefore reducing interpretability for certain time periods of the dataset.

While this study was able to achieve its primary objectives, there are several areas that could be improved to enhance the interpretability of the findings that were obtained. For example, by incorporating embeddings that consider sarcastic or ironic language (e.g. BERT word embedding models), The sentiment scores produce from the YouTube corpus could better reflect the contextual environment that language is in used online. Additionally, the study did not attempt to categorise users into communities to identify the types of groups that were driving semantic change. So, if the priority of user anonymisation was no longer a factor, the study could be developed to provide insights into how language is recontextualised in different online communities.

Appendix

```python
# this fucntion is based on the YouTube Data API v3 documentation
# https://developers.google.com/youtube/v3/docs/search/list
def search_videos(keyword, year):
    """
    This is a function to request to collect the Metadata for a given YouTube video using Youtube Data API.

    Parameters:
    1. the keyword to search for in the video titles and descriptions.
    2. the target year to limit the video search to (2010-2025).

    Returns:
    1. a dictionary of the metadata for the videos that match the search criteria.
    """
    results = youtube.search().list(
        part = 'snippet',
        maxResults = 100, # the maximum number of videos to retrieve in the request
        publishedAfter = f'{year}-01-01T00:00:00Z',
        publishedBefore = f'{year}-12-31T23:59:59Z',
        q = keyword, # the keyword to search for (e.g. woke)
        relevanceLanguage = 'en', # search is limited to only English content
        type = 'video' # the search query is only restricted to videos (excl. playlists, channels)
    )
    return results.execute()
```

***Appendix 1**: Keyword search for YouTube metadata*

```python
def fetch_top_level_comments(video_id):
    """
    This is a fucntion to call all the top-level comments for a given Youtube video using the Youtube Data API

    Parameters:
    1. video_id (str): the uniqiue identifier of the Youtube videos.

    Returns:
    1. a list of plain-text comment strings to use in sentiment and semantic drift analysis.
    """

    top_level_comments = [] # list to store all extracted "top-level" comments
    try:
        results = youtube.commentThreads().list(
                part ='snippet',
                videoId = video_id,
                maxResults = 10, # retrieving up to 10 top-level comments
                textFormat = 'plainText'
        )
        response = results.execute()

        for item in response.get('items', []):
            comment_text = item['snippet']['topLevelComment']['snippet']['textDisplay']
            top_level_comments.append(comment_text)

    except Exception as e:
        print(f"Unexpected error fetching comments for video {video_id}: {e}")

    return top_level_comments
```

***Appendix 2**: Function to fetch top level comments of YouTube metadata*

```python
def clean_text(text):
    """
    This function cleans and normalsises the data for Tokenisation and preprocessing
    """
    # removes all lowercase, special characters and whitespace
    text = text.lower()
    text = re.sub(r'[a^zA-Z\s]', '', text)
    text = re.sub(r'[\U00010000-\U0010ffff]', '', text) # remove emojis
    text = text.strip() # remove whitespace

    # tokenize text and filter stopwords out of text
    tokens = word_tokenize(text)
    tokens = [t for t in tokens if t not in stop_words and t.isalpha()] # remove the noise

    # cleaned sentence string
    clean_string = ' '.join(tokens) # for sentiment analysis
    return tokens, clean_string
```

Appendix 3: *Normalisation Function*

```python
def compound_score_classification(comp_score):
    if comp_score >= 0.05:
        return 'pos'
    elif comp_score <= -0.05:
        return 'neg'
    else:
        return 'neu'
```

Appendix 4: *Compound Score Classification Function*

```python
year_embeddings = {}

# extracting all instances of 'woke' per year
for year, sentences in tokenised_by_year.items():
    vectors = []

    # looping through each sentence in the year and each token in the sentence
    for sentence in sentences:
        if isinstance(sentence, tuple):
            sentence = sentence[0]

        doc = nlp("  ".join(sentence)) # convert token to a vector using spaCy
        for token in doc:
                  if token.text.lower() == target_word:
                      vectors.append(token.vector)

    if vectors:
        year_embeddings[year] = np.mean(vectors, axis=0)
        print(f"{year}: {len(vectors)} vectors used")
    else:
        print(f"{year}: '{target_word}' not found. Skipping")
```

Appendix 5: *Word embeddings calculation*

```python
# Cosine similarty drift from 2010 baseline
base_year = 2010

if base_year not in year_embeddings:
    print(f"No embedding found for base year {base_year}. Skipping calculation")
else:
    base_vector = year_embeddings[base_year]

    semantic_drift = {}

    for year, vector in year_embeddings.items():
        similarity = cosine_similarity([base_vector], [vector])[0][0]
        cosine_distance = 1-similarity
        semantic_drift[year] = round(cosine_distance, 4)

 #Save to DataFrame
sema_drift_df = pd.DataFrame({
    'Year': list(semantic_drift.keys()),
    'Semantic Drift from 2010': list(semantic_drift.values())
}).sort_values('Year')

#Save to csv
sema_drift_df.to_csv("Youtube_Semantic_Drift_from_2010.csv", index=False)
```

Appendix 6: *Cosine similarity calculation*

```python
def get_cooccurrence(sentences, window_size=10): # window size of 10 positions of each other

    cooccurrences = Counter()
    for tokens in sentences:
        for i, token in enumerate(tokens):
            for j in range(i+1, min(i + window_size + 1, len(tokens))):
                w1, w2 = tokens[i], tokens[j]
                if w1 == w2:
                    continue
                if w1 in string.punctuation or w2 in string.punctuation:
                    continue
                if w1 in stop_words or w2 in stop_words:
                    continue

                pair = tuple(sorted([w1, w2]))
                cooccurrences[pair] += 1
    return cooccurrences
```

***Appendix** 7: Co-occurrence Function*

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